

Causal Inference Methods for Infectious Disease Prevention Trials

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March 25, 2026



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Outline

- **Project 2:** Physical Distancing to Reduce Transmission of Influenza-Like-Illness on College Campuses: the eX-FLU Trial

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- **Project 2:** Physical Distancing to Reduce Transmission of Influenza-Like-Illness on College Campuses: the eX-FLU Trial
- **Project 3:** Causal Inference from Cluster Randomized Trials with Differential Nonresponse
- **Project 1:** Addressing Confounding and Continuous Exposure Measurement Error Using Corrected Score Functions

Project 2: Physical Distancing to Reduce Transmission of Influenza-Like-Illness on College Campuses: the eX-FLU Trial

Brian Richardson, Allison Aiello, Michael Hudgens

- **eX-FLU**: trial to evaluate a social distancing intervention on a college campus during flu season ([Aiello et al., 2016](#); [Zivich et al., 2020](#))

Design and methods of a social network isolation study for reducing respiratory infection transmission: The eX-FLU cluster randomized trial

Allison E. Aiello^{a,*}, Amanda M. Simanek^{b,1}, Marisa C. Eisenberg^c, Alison R. Walsh^c, Brian Davis^c, Erik Volz^{d,1}, Caroline Cheng^c, Jeanette J. Rainey^e, Amra Uzicanin^e, Hongjiang Gao^e, Nathaniel Osgood^f, Dylan Knowles^f, Kevin Stanley^f, Kara Tarter^c, Arnold S. Monto^c

eX-FLU Trial

- **eX-FLU**: trial to evaluate a social distancing intervention on a college campus during flu season ([Aiello et al., 2016](#); [Zivich et al., 2020](#))
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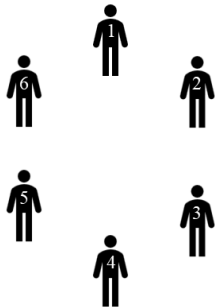
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- **Intervention**: encouragement to isolate in dorm for three days upon developing symptoms of **influenza-like illness** (ILI)
- **Central question**: does the encouragement-to-isolate intervention reduce transmission of ILI?

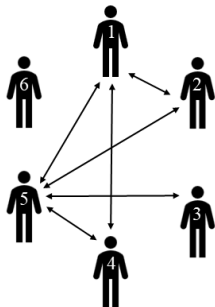
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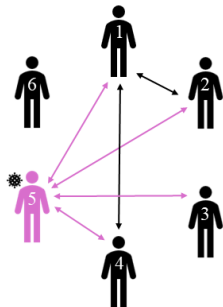
Transmission of Influenza-Like-Illness



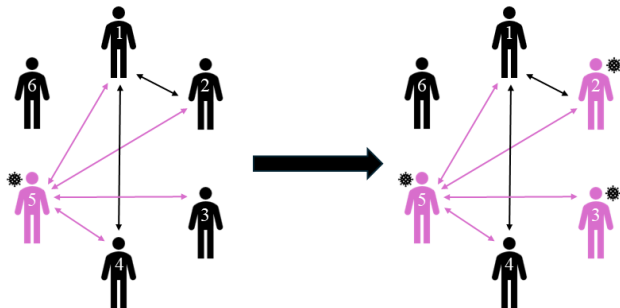
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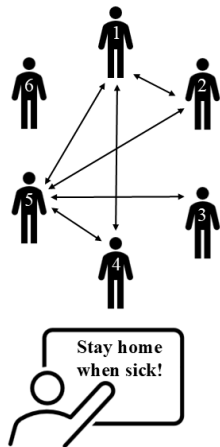
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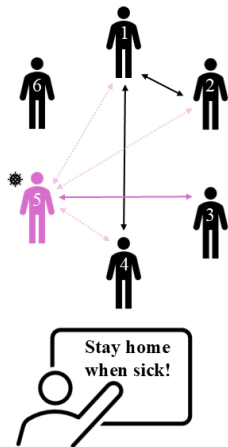
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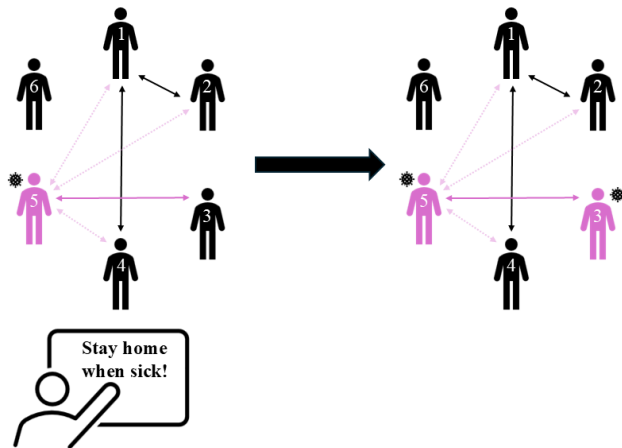
Example I: Intervention Affects Network and Transmission



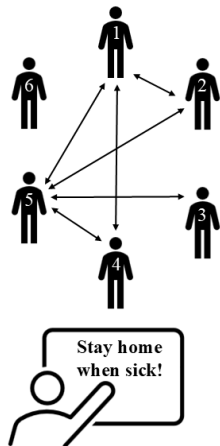
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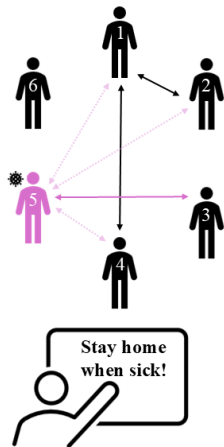
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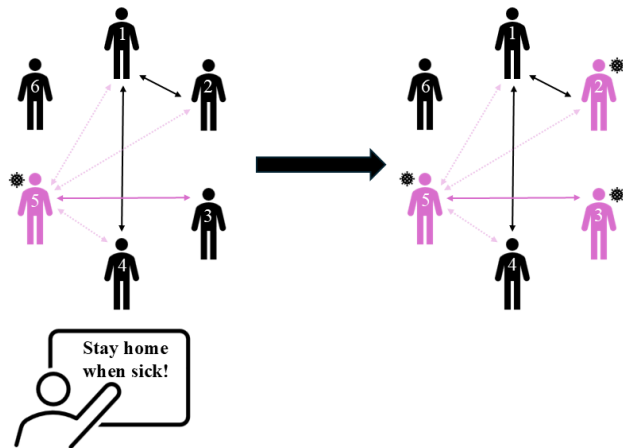
Example II: Intervention Affects Network, Not Transmission



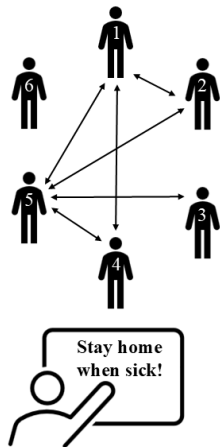
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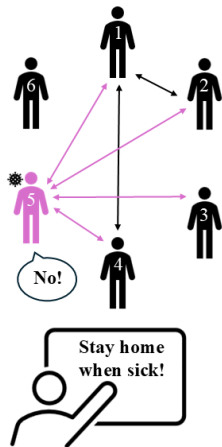
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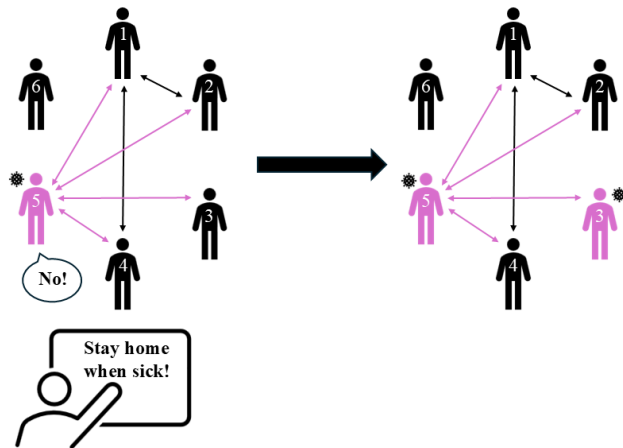
Example III: Intervention Affects Transmission, Not Network



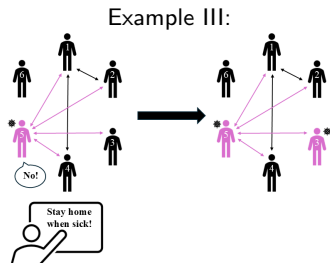
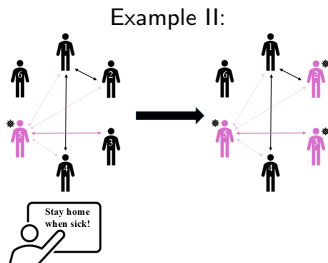
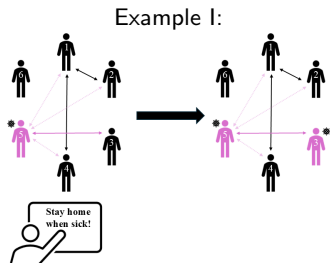
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Central Question: Does the Intervention Affect Transmission of ILI?



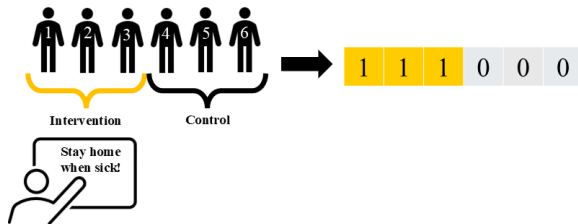
In Examples I and III, the answer is yes

eX-FLU Observed Data

- Baseline randomization assignments

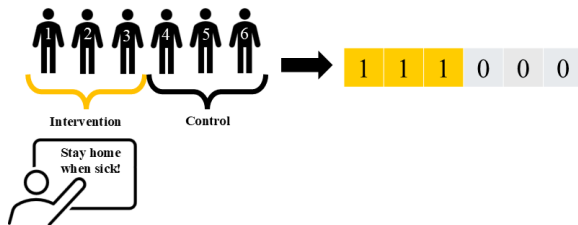
$\mathbf{Z} = (Z_1, \dots, Z_n) \in \mathcal{Z}$, for

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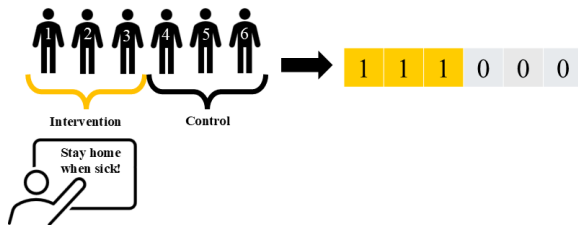
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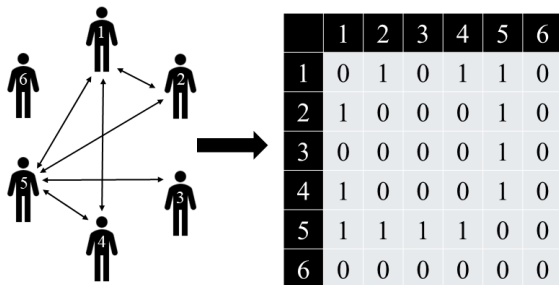
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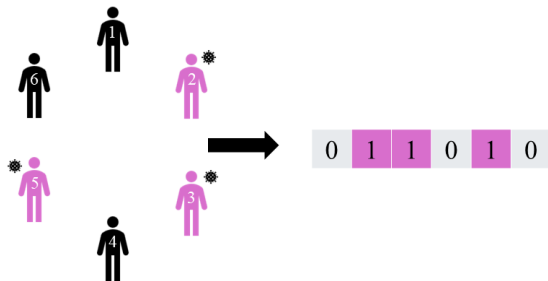
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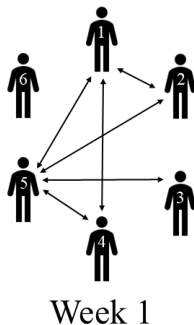
- ▶ infections $\mathbf{Y}^k = (Y_1^k, \dots, Y_n^k)$, for

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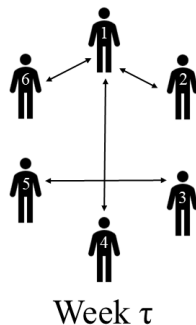


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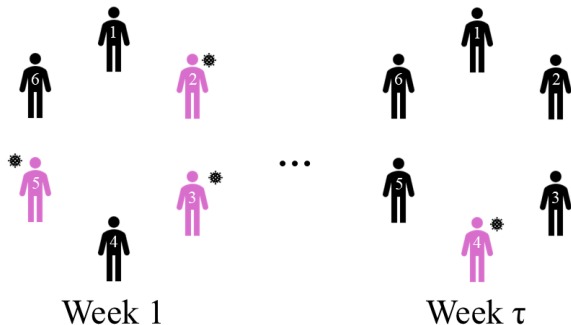


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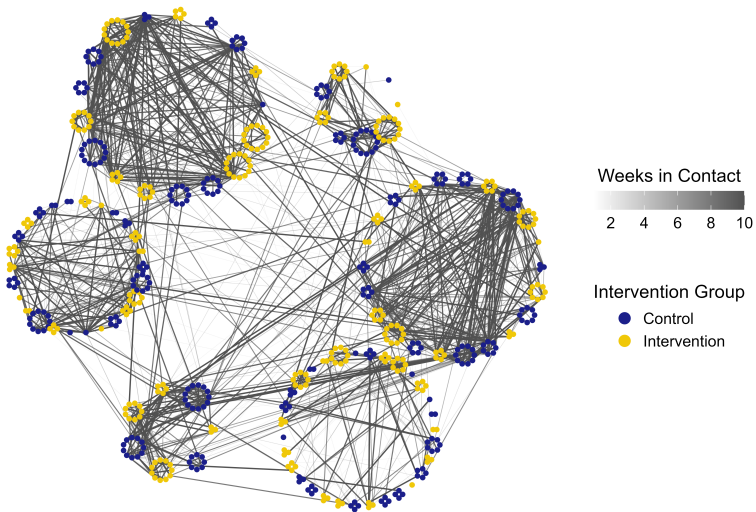


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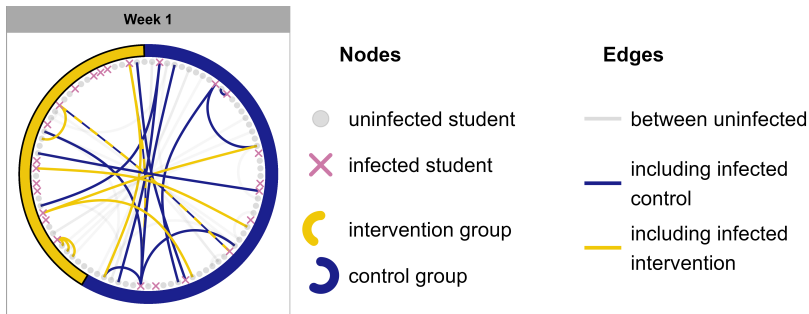
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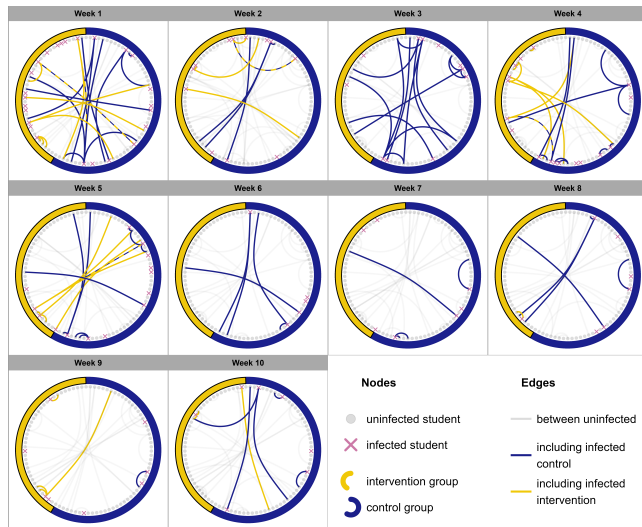


eX-FLU Observed Data



(93 out of 579 students with at least one infection)

eX-FLU Observed Data



eX-FLU Potential Outcomes

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Potential outcomes are defined for the networks and ILI infections:

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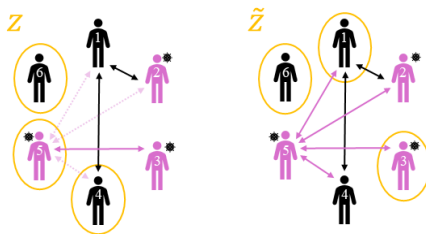
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([Hudgens and Halloran, 2008](#))
- Assume **causal consistency**:

$$\overline{\mathbf{A}} = \overline{\mathbf{A}}(\mathbf{Z}) \text{ and } \overline{\mathbf{Y}} = \overline{\mathbf{Y}}(\mathbf{Z})$$

eX-FLU Null Hypotheses

$$H_0^Y : \bar{Y}(\mathbf{z}) = \bar{Y}(\tilde{\mathbf{z}}) \text{ for all } \mathbf{z}, \tilde{\mathbf{z}} \in \mathcal{Z}$$

(“the intervention has no effect on the infections”)



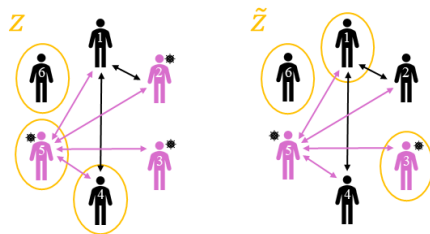
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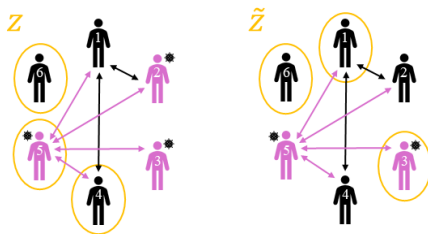
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$$H_0^A : \bar{A}(z) = \bar{A}(\tilde{z}) \text{ for all } z, \tilde{z} \in \mathcal{Z}$$

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$$H_0^\sharp = H_0^Y \cap H_0^A : \bar{Y}(z) = \bar{Y}(\tilde{z}) \text{ and } \bar{A}(z) = \bar{A}(\tilde{z}) \text{ for all } z, \tilde{z} \in \mathcal{Z}$$

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Previous analyses of the eX-FLU trial ([Alexandria et al., 2023](#)) tested $H_0^\sharp$, but no analyses have tested H_0^Y

Testing $H_0^\#$

“How unlikely are the observed data under $H_0^\#$?”

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Randomization	Networks	Infections	Test Statistic
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$\mathbf{z}_2$			
$\dots$			
$\mathbf{z}_{ \mathcal{Z} }$			

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$\dots$	$\dots$	$\dots$	$\dots$
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...	...	...	...
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$\mathbf{z}_1$	$\bar{\mathbf{A}}(\mathbf{z}_1) = \bar{\mathbf{A}}$	$\bar{\mathbf{Y}}(\mathbf{z}_1) = \bar{\mathbf{Y}}$	$T(\mathbf{z}_1, \bar{\mathbf{A}}, \bar{\mathbf{Y}})$
$\mathbf{z}_2$	$\bar{\mathbf{A}}(\mathbf{z}_2) = \bar{\mathbf{A}}$	$\bar{\mathbf{Y}}(\mathbf{z}_2) = \bar{\mathbf{Y}}$	$T(\mathbf{z}_2, \bar{\mathbf{A}}, \bar{\mathbf{Y}})$
...	...	...	...
$\mathbf{z}_{ \mathcal{Z} }$	$\bar{\mathbf{A}}(\mathbf{z}_{ \mathcal{Z} }) = \bar{\mathbf{A}}$	$\bar{\mathbf{Y}}(\mathbf{z}_{ \mathcal{Z} }) = \bar{\mathbf{Y}}$	$T(\mathbf{z}_{ \mathcal{Z} }, \bar{\mathbf{A}}, \bar{\mathbf{Y}})$

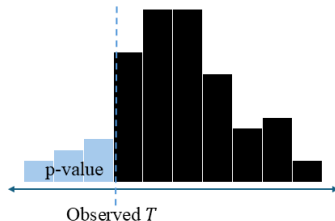
Testing $H_0^\sharp$

“How unlikely are the observed data under $H_0^\sharp$?” We can answer this using a **randomization test** (Fisher, 1936; Alexandria et al., 2023; Zhang and Zhao, 2023; Ritzwoller et al., 2024)

- Choose a test statistic $T = T(\mathbf{Z}, \bar{\mathbf{A}}, \bar{\mathbf{Y}})$
- Find the distribution of T under $H_0^\sharp$
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- Compute a p-value: $\rho^\sharp = F_T(T|H_0^\sharp)$, where

$$F_T(t|H_0^\sharp) = \Pr(T \leq t|H_0^\sharp) = \sum_{\mathbf{z} \in \mathcal{Z}} r(\mathbf{z}) \mathbb{1}\{T(\mathbf{z}, \bar{\mathbf{A}}, \bar{\mathbf{Y}}) \leq t\}$$

is the **cumulative distribution function** (CDF) of T



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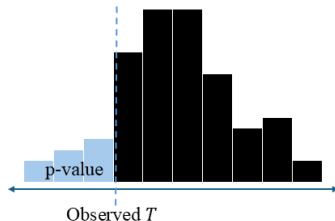
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is the **cumulative distribution function** (CDF) of T

- Reject $H_0^\sharp$ if $\rho^\sharp \leq 0.05$



Testing $H_0^\sharp$

- This test of $H_0^\sharp$ controls the **type I error rate** exactly:

$$\Pr(\rho^\sharp \leq \alpha | H_0^\sharp) \leq \alpha$$

for any $\alpha \in [0, 1]$, for any sample size, and for any choice of test statistic

Testing $H_0^\sharp$

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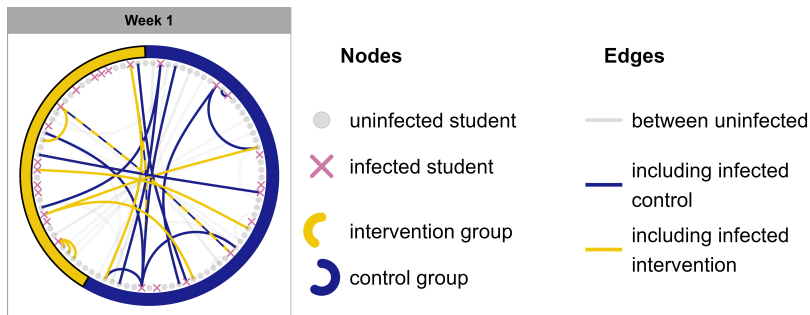
- The **power** of this test:

$$\Pr(\rho^\sharp \leq \alpha | H_1^\sharp)$$

depends on the choice of test statistic

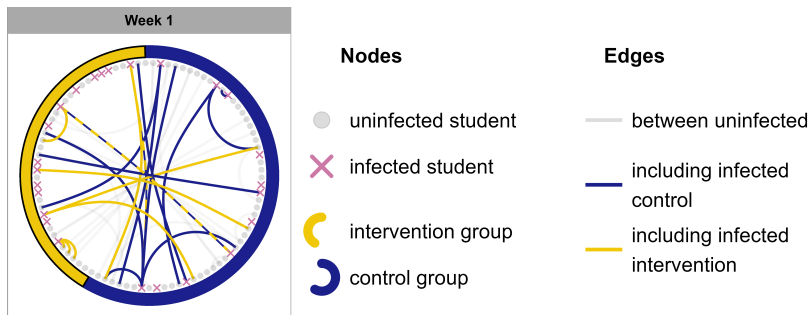
Choice of Test Statistic

- 93 (out of 579) students with at least one infection



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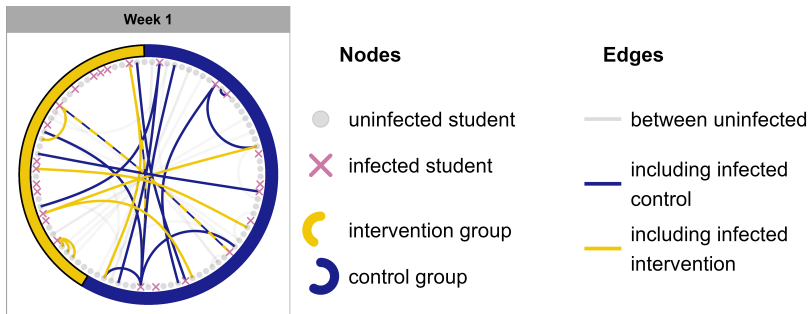
- 93 (out of 579) students with at least one infection
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Choice of Test Statistic

- 93 (out of 579) students with at least one infection
- Bold edges represent possible transmission events
 - ▶ Proportion of possible transmission events attributable to students in the intervention group:

$$T = \frac{\text{number of yellow edges}}{\text{total number of edges}} = \frac{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j \neq i} Z_i Y_i^{k-1} A_{ij}^{k-1}}{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j \neq i} Y_i^{k-1} A_{ij}^{k-1}} = 0.359$$



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Testing H_0^Y

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- 1 Estimate the unknown parameter θ in the PMF $q(\bar{\mathbf{a}}, \mathbf{z}; \theta)$ of $\bar{\mathbf{A}}(\mathbf{z})$

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$$\hat{\rho}_B^Y = B^{-1} \sum_{b=1}^B \mathbb{1}(T_b \leq T)$$

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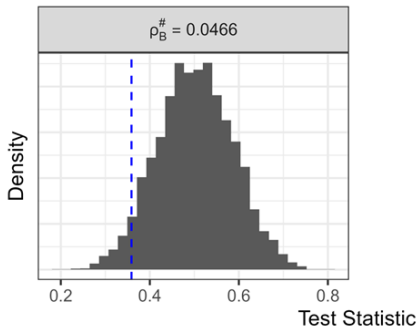
Proposition: This procedure will **asymptotically** control the type I error rate

eX-FLU: Hypothesis Test Results

- Test statistic (proportion of possible transmission events attributable to students in the intervention group):
 $T = 0.359$.

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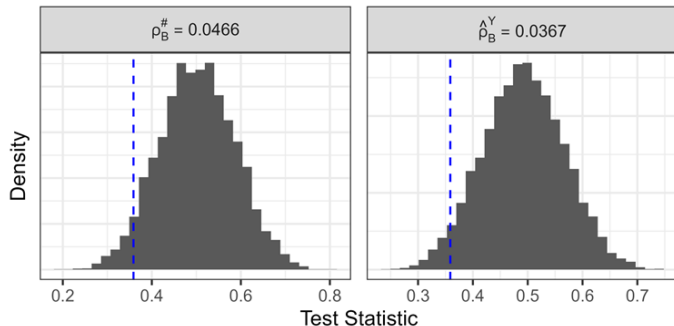
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- Encouragement to isolate affects the social network and/or influenza-like illness ($\rho_B^\# = 0.0466$)
- Encouragement to isolate specifically affects influenza-like illness ($\hat{\rho}_B^Y = 0.0367$)

Future Directions

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 - ▶ trade-off between robustness and utility
- ③ **Design future trials** with independent clusters
 - ▶ can allow **identification** of more causal estimands

Summary of Project 2

- Developed a new randomization-based inference procedure
 - ▶ Established theoretical properties (asymptotic control of Type I error)
 - ▶ Demonstrated empirical performance through simulations
- Applied the method to the **eX-FLU trial** to find a protective effect of an encouragement-to-isolate intervention on a college campus
- **Status:** under revision at *Proceedings of the National Academy of Sciences*

Project 3 : Causal Inference from Cluster Randomized Trials with Differential Nonresponse

Brian Richardson, Bonnie Shook-Sa, Michael Hudgens

PopART Trial

- **PopART**: large **cluster randomized trial** designed to evaluate a combination **HIV prevention intervention** in Zambia and South Africa ([Hayes et al., 2014](#))

Hayes et al. *Trials* 2014, **15**:57
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STUDY PROTOCOL

Open Access

HPTN 071 (PopART): Rationale and design of a cluster-randomised trial of the population impact of an HIV combination prevention intervention including universal testing and treatment – a study protocol for a cluster randomised trial

Richard Hayes¹, Helen Ayles^{2,3}, Nulda Beyers⁴, Kalpana Sabapathy^{1*}, Sian Floyd¹, Kwame Shanaube³, Peter Bock⁴, Sam Griffith⁵, Ayana Moore⁵, Deborah Watson-Jones², Christophe Fraser⁶, Sten H Vermund⁷, Sarah Fidler⁸ and The HPTN 071 (PopART) Study Team

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 - ▶ Arm A: annual home-based HIV testing, promotion of medical male circumcision for HIV-negative men, and offer of immediate ART for those testing HIV-positive
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- **Outcome**: HIV Incidence

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PopART Trial

- **Initial Findings:** no protective effect of the Arm A intervention ([Hayes et al., 2019](#))
 - ▶ “unanticipated and not consistent with the data on viral suppression”



Effect of Universal Testing and Treatment on HIV Incidence — HPTN 071 (PopART)

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PopART Trial

- **Initial Findings:** no protective effect of the Arm A intervention ([Hayes et al., 2019](#))
 - ▶ “unanticipated and not consistent with the data on viral suppression”
- **Later Analyses:** identified possibly differential nonresponse that could bias trial results ([Shook-Sa et al., 2025](#))
 - ▶ Males were less likely to respond (i.e., have data collected)
 - ▶ The intervention appeared more effective among males



Effect of Universal Testing and Treatment on HIV Incidence — HPTN 071 (PopART)

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Goal

To estimate the **causal effect** of a cluster-level exposure on an outcome under:

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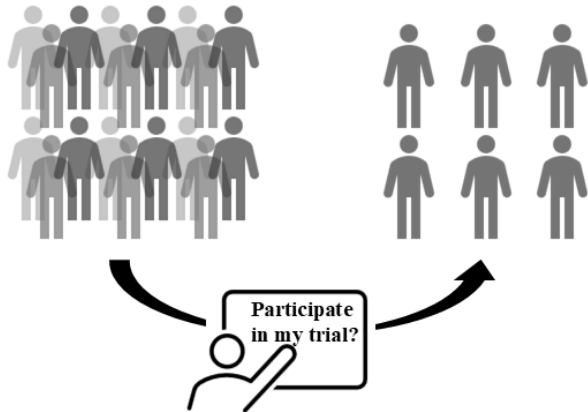
To estimate the **causal effect** of a cluster-level exposure on an outcome under:

- (i) individual-level **nonresponse** that may depend on missing covariates and on treatment
- (ii) outcome **censoring**

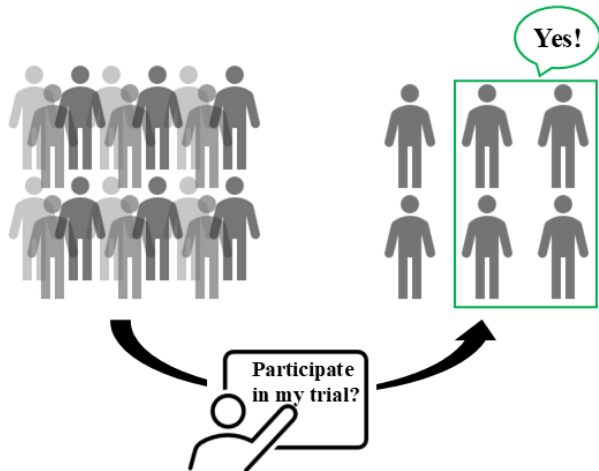
(Non)Participation



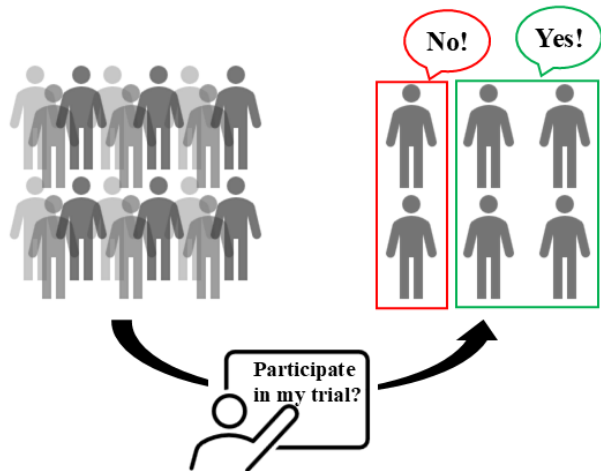
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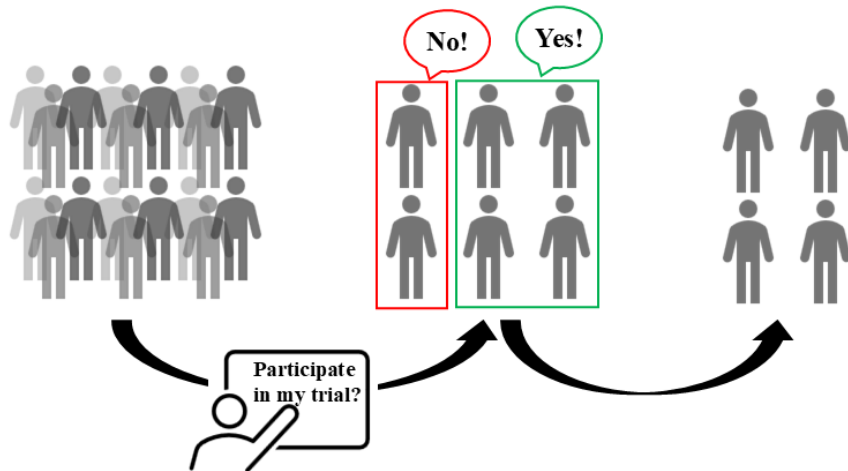
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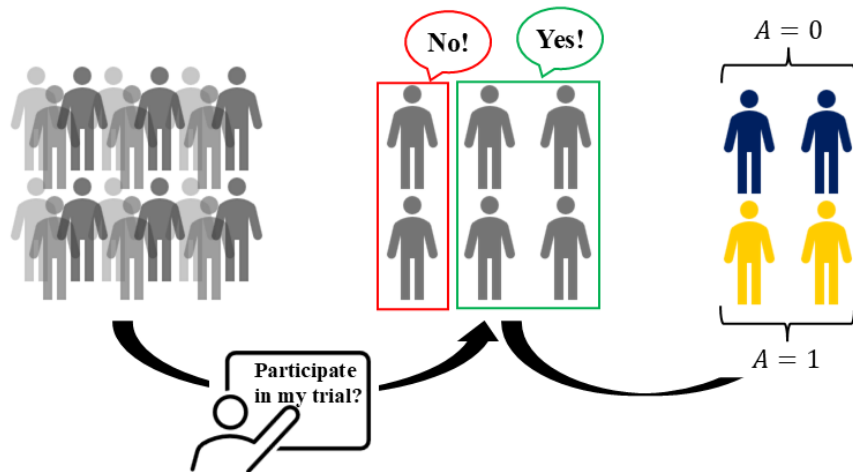
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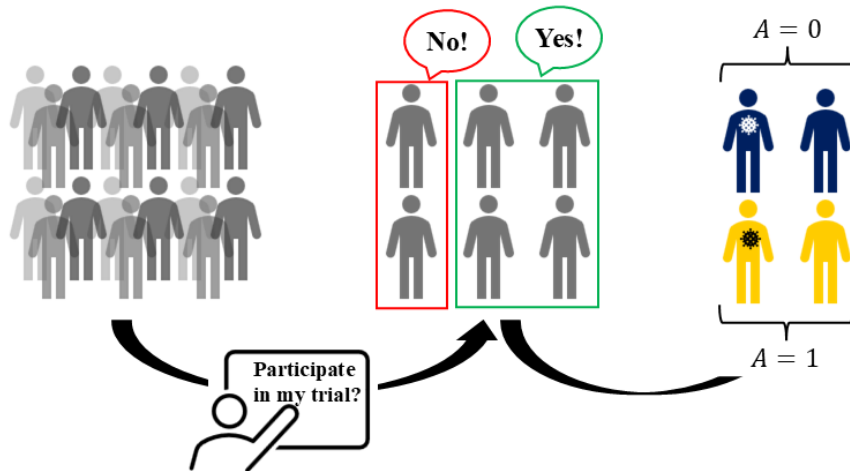
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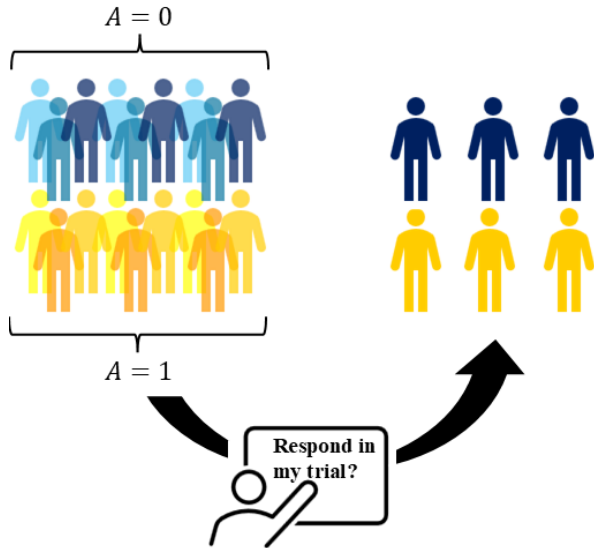
(Non)Response



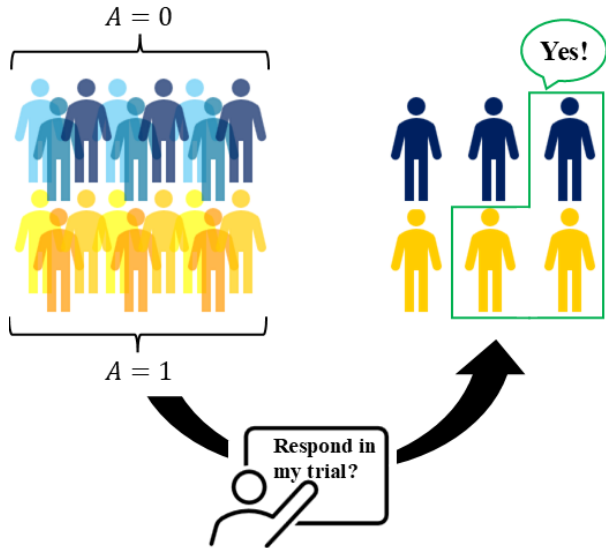
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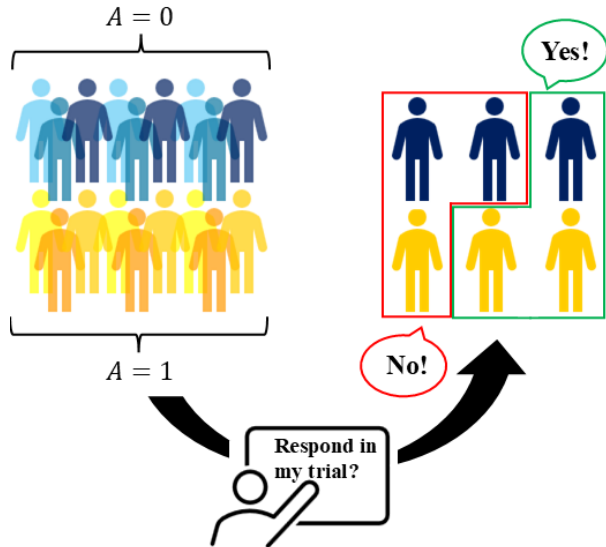
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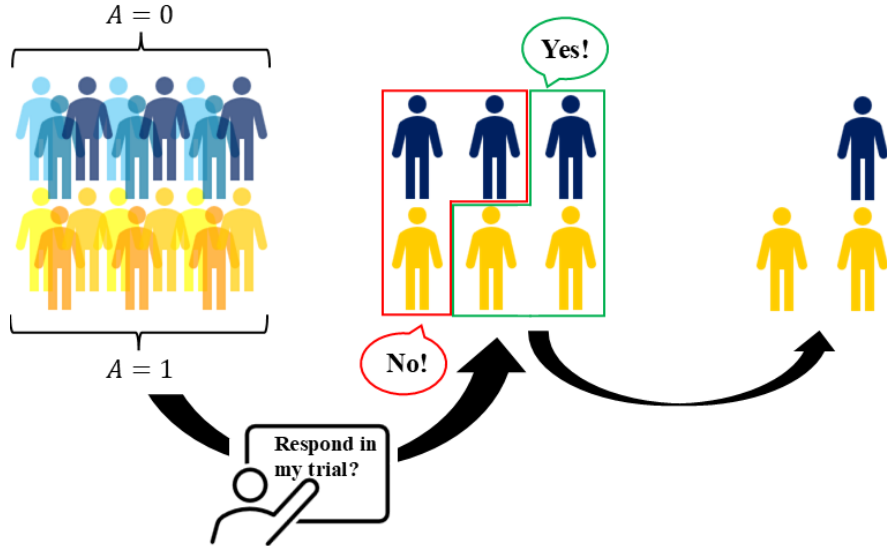
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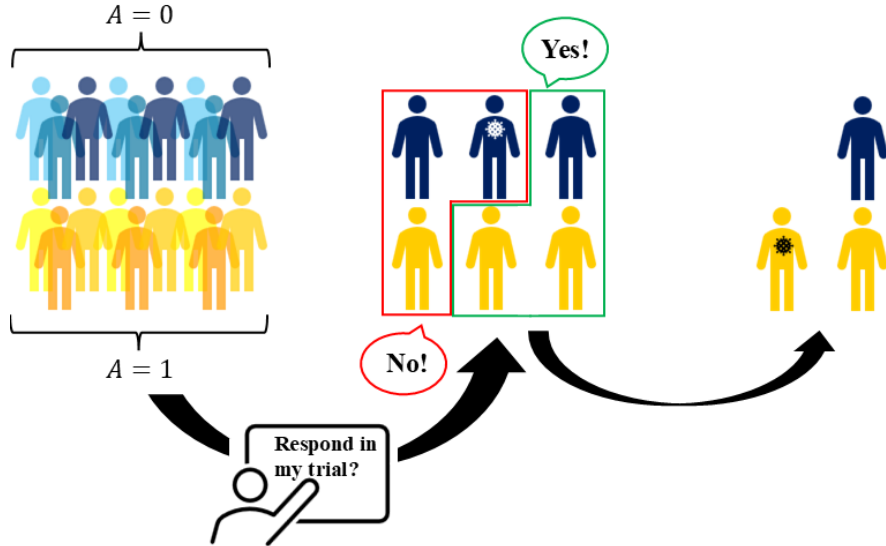
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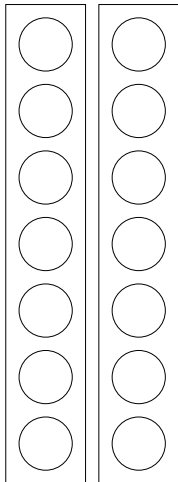


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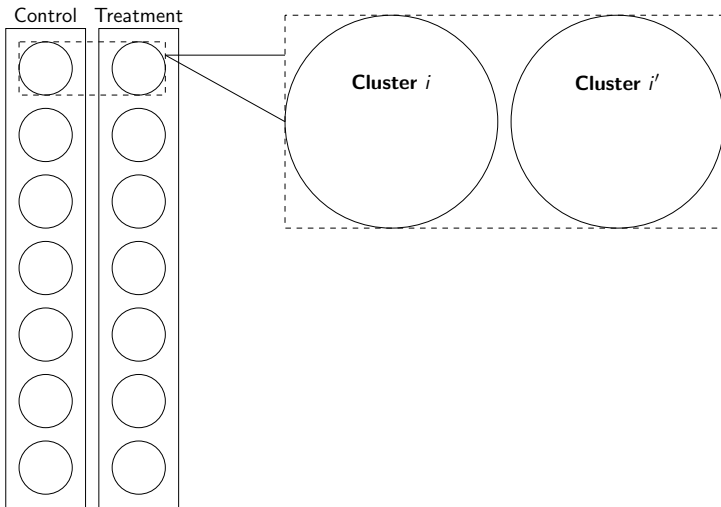


PopART Observed Data

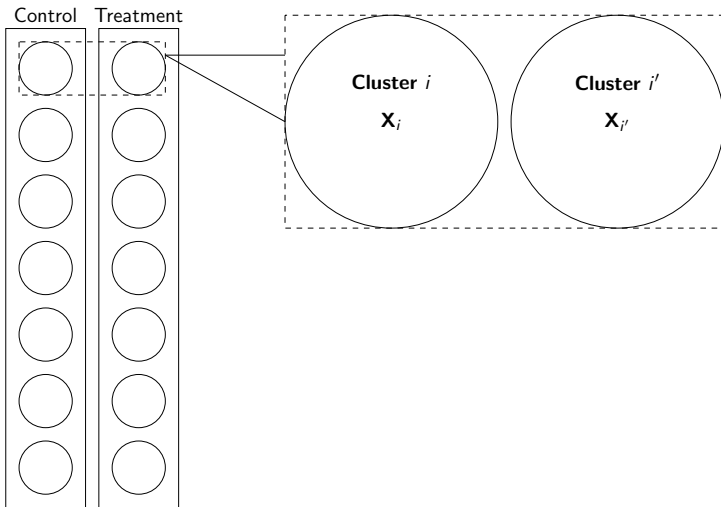
Control Treatment



PopART Observed Data

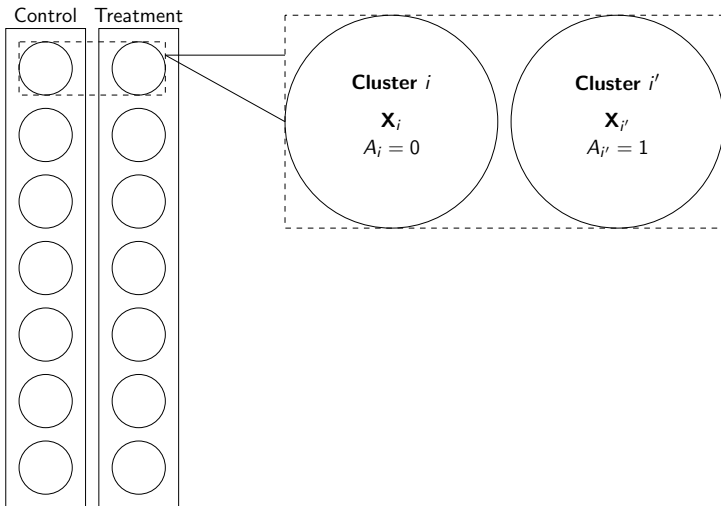


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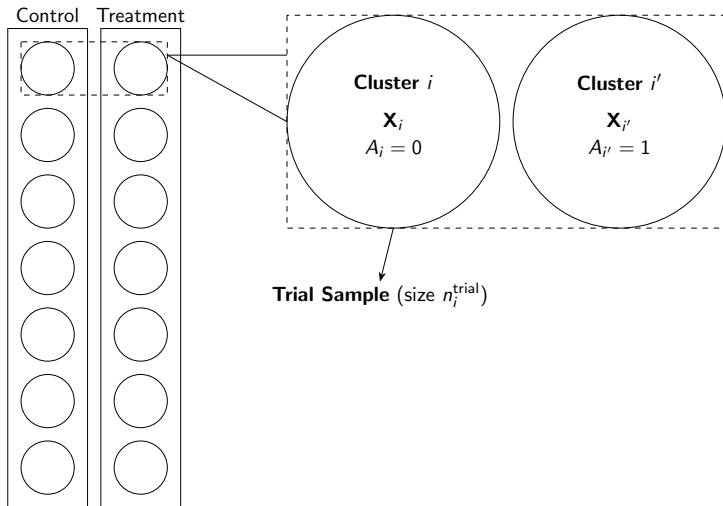
- X_i : cluster covariates

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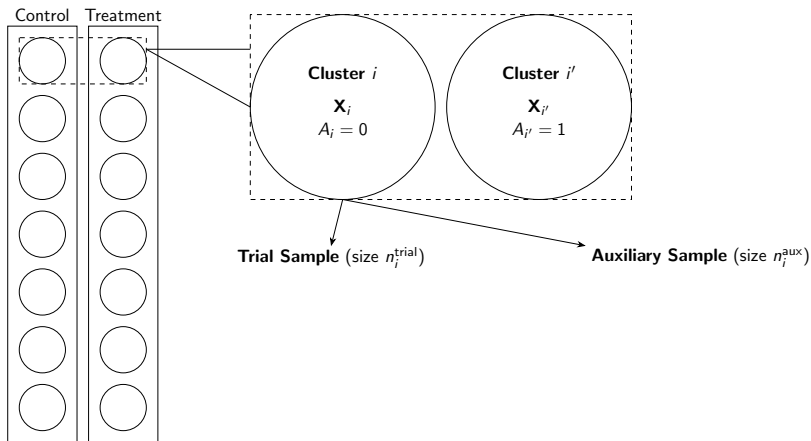
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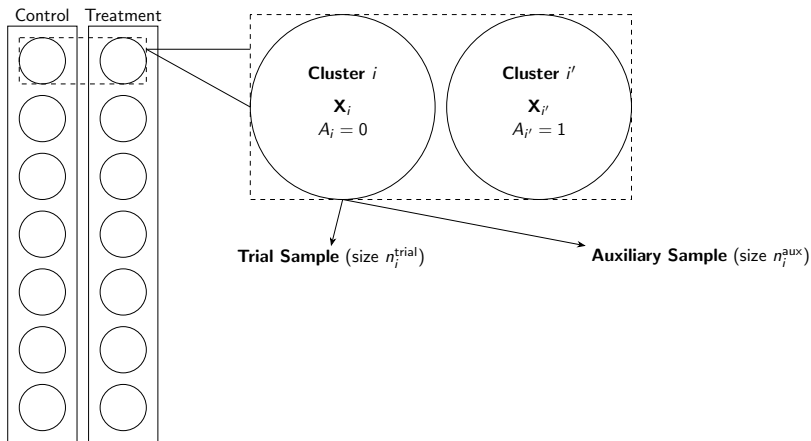
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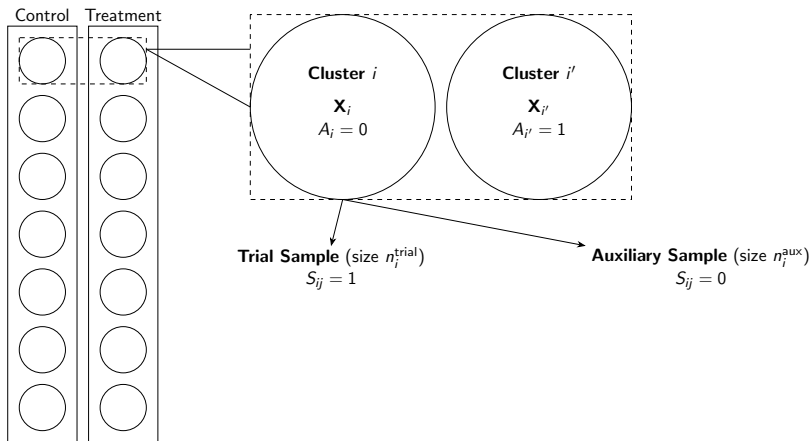
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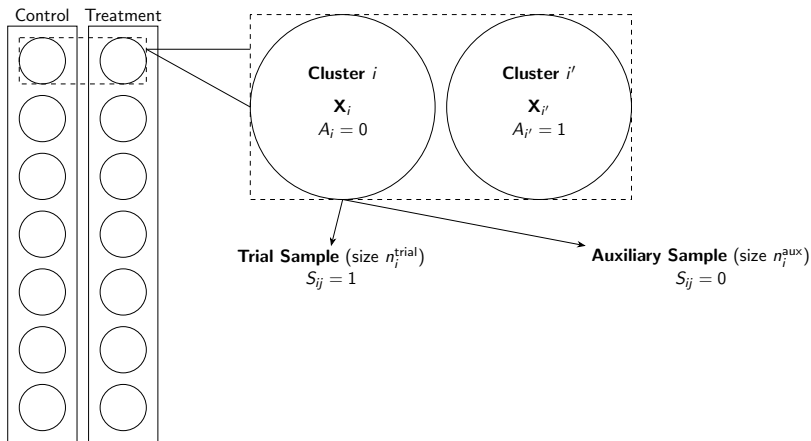
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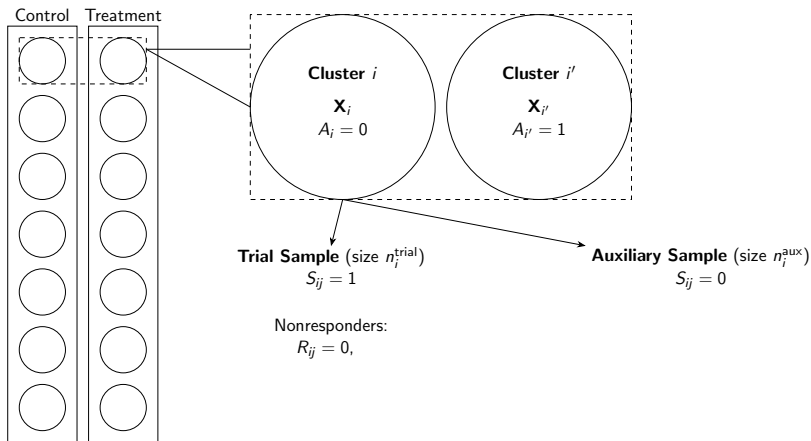
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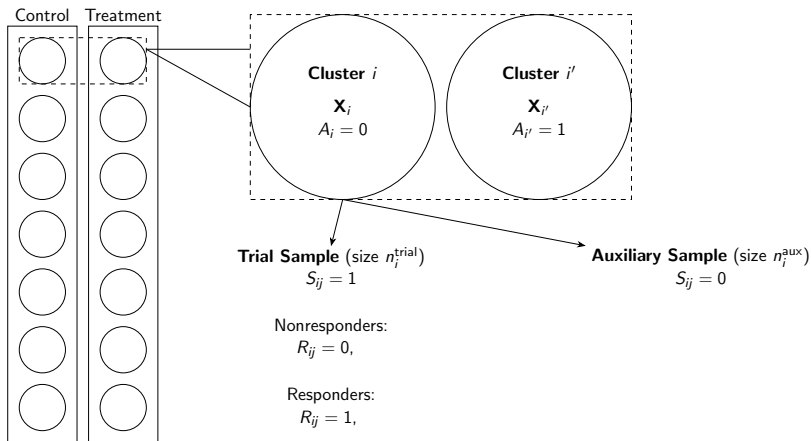
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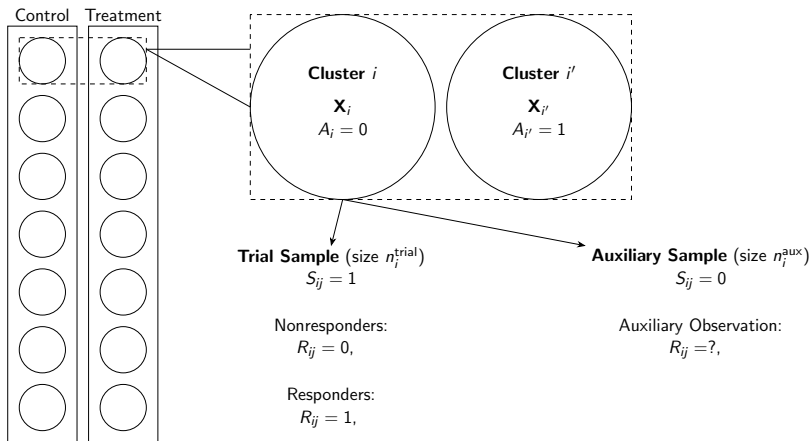
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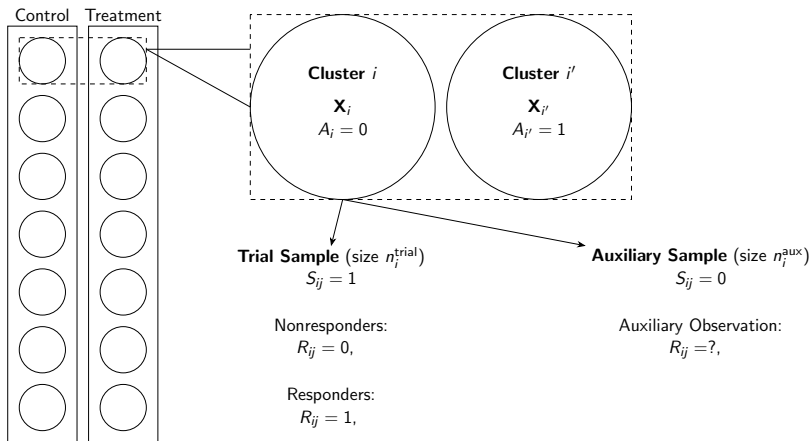
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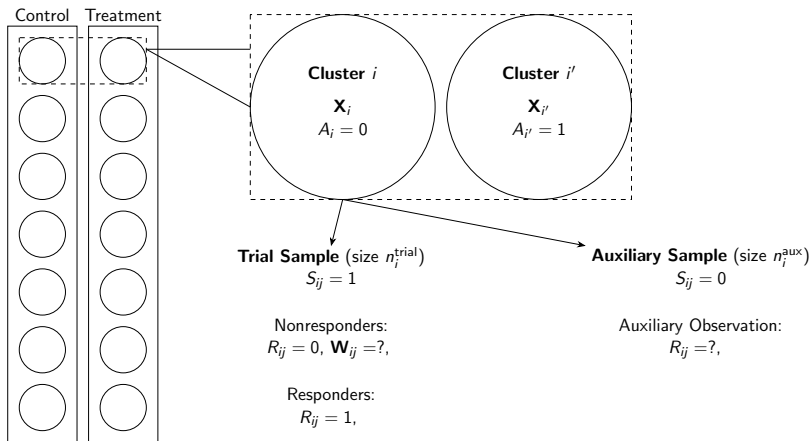
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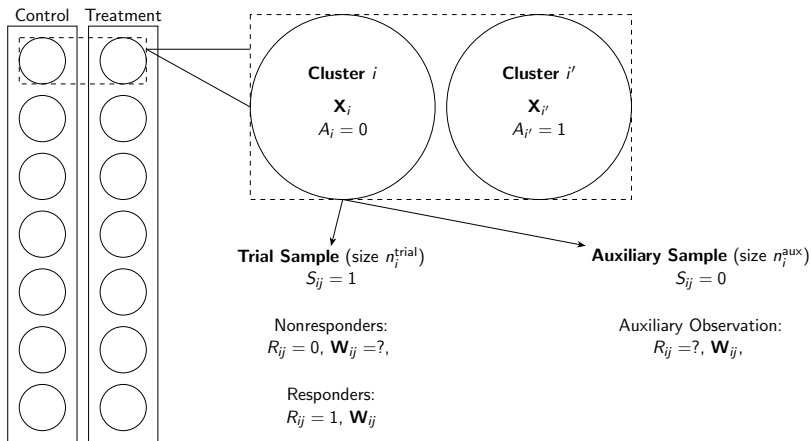
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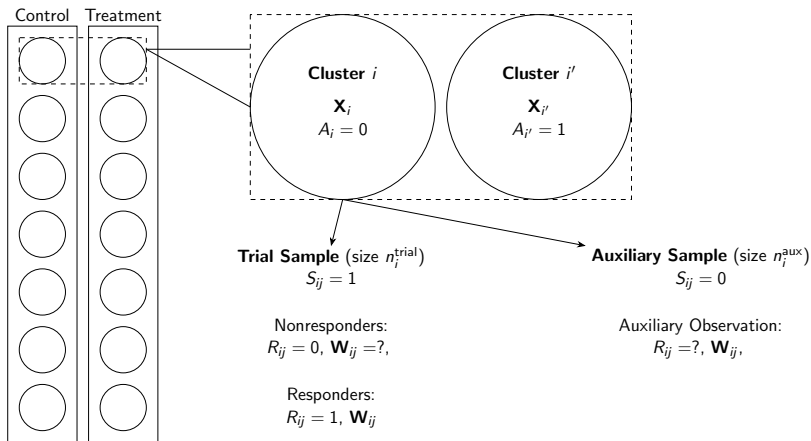
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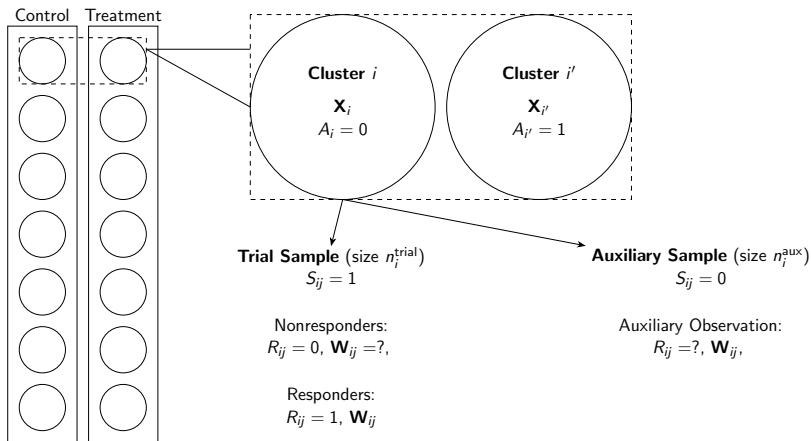
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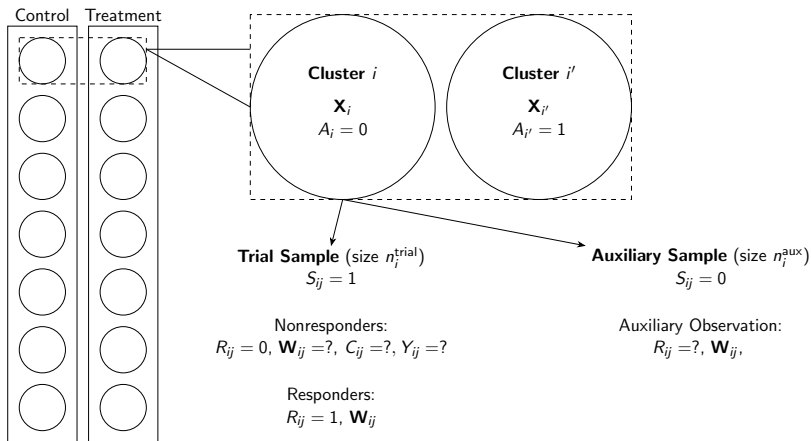
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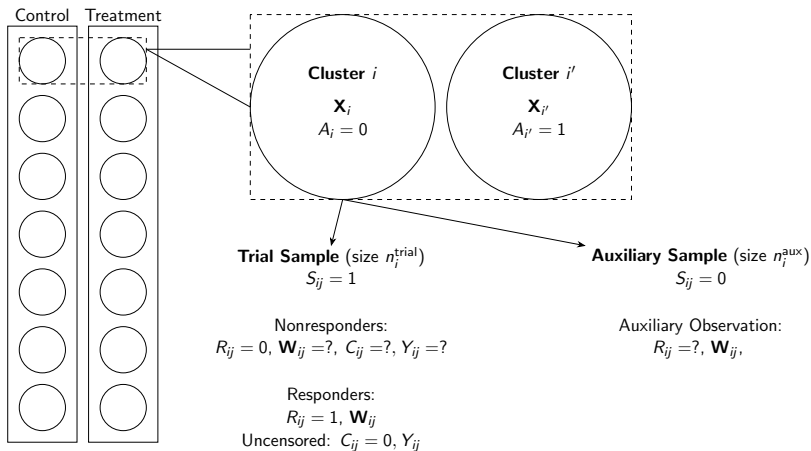
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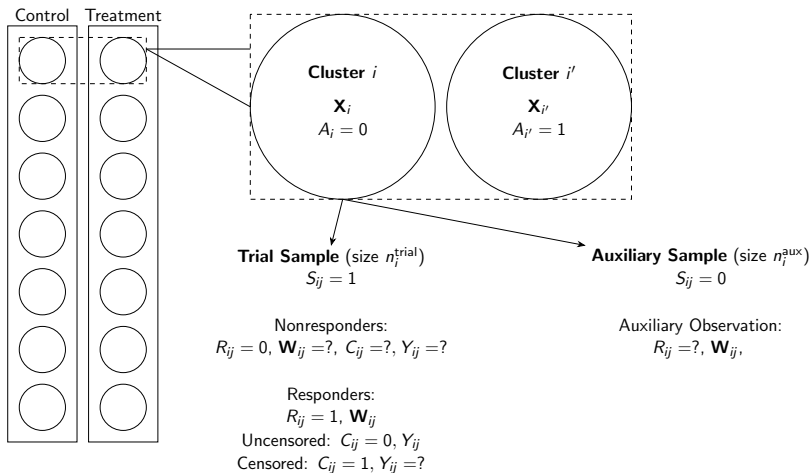
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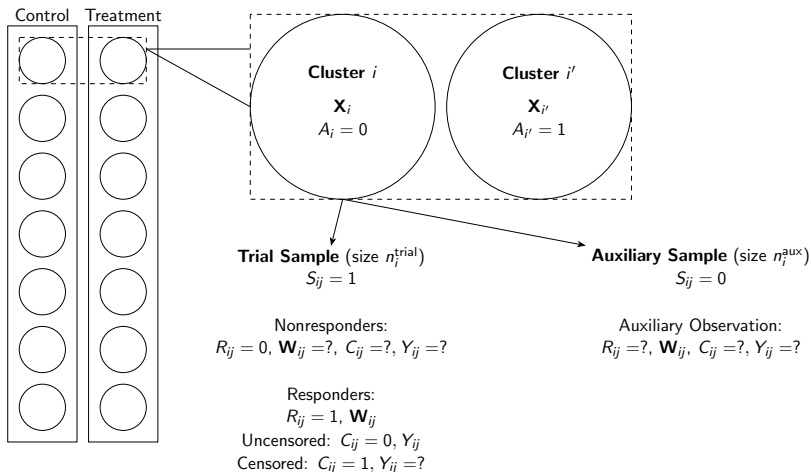
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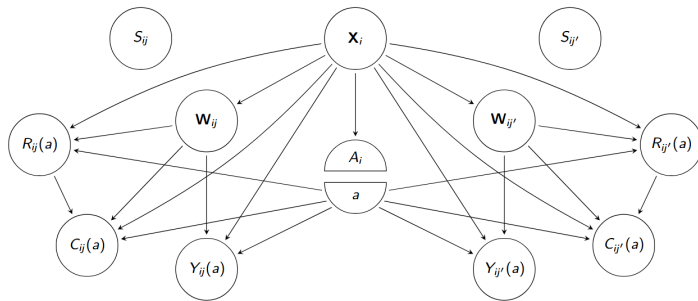
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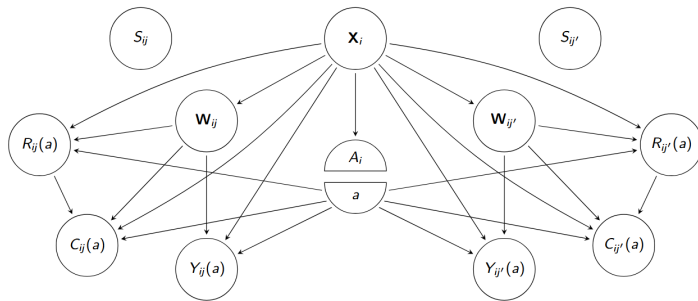
$$RD = \eta(1) - \eta(0), \quad RR = \frac{\eta(1)}{\eta(0)}, \quad OR = \frac{\eta(1)/\{1 - \eta(1)\}}{\eta(0)/\{1 - \eta(0)\}}$$

Assumptions (Briefly)



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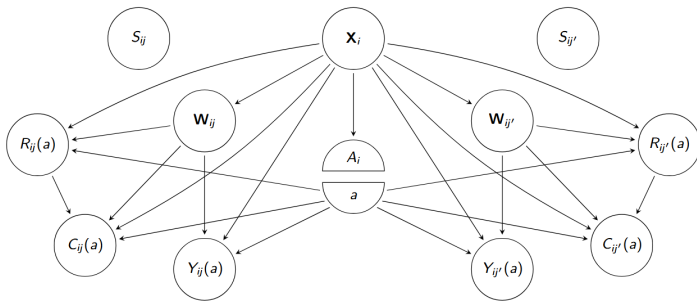
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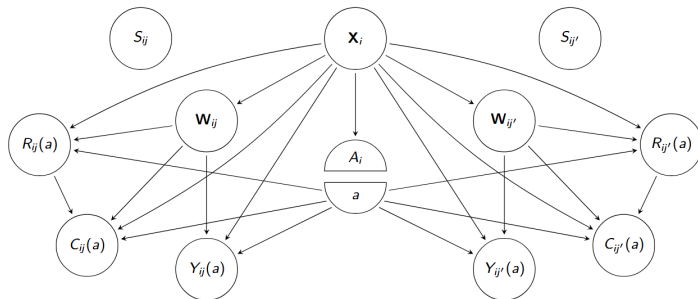
- ▶ **Cluster-randomization:** $A_i \perp\!\!\!\perp (\mathbf{W}_{ij}, \dots, \mathbf{W}_{in_i}) | \mathbf{X}_i$



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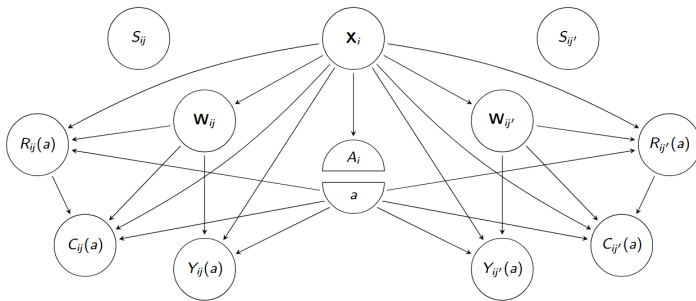
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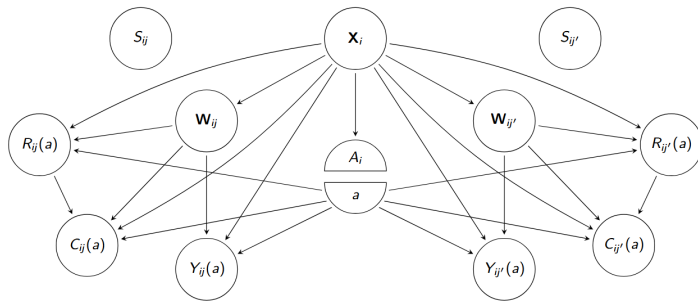
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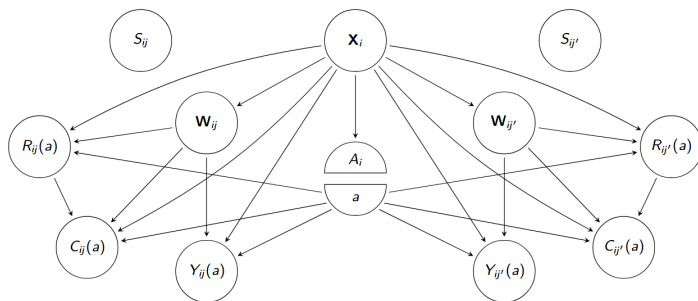
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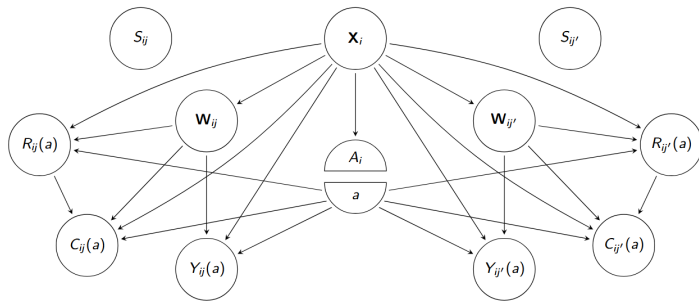
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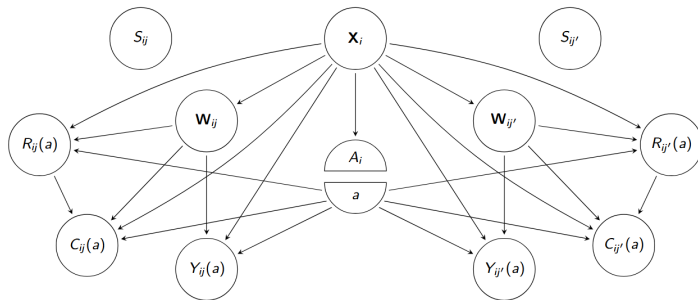
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 - ▶ **Positivity of treatment assignment:** $\Pr(A_i = a | \mathbf{X}_i = \mathbf{x}) > 0 \forall \mathbf{x}$ with positive density $f_{\mathbf{X}}(\mathbf{x}) > 0$.



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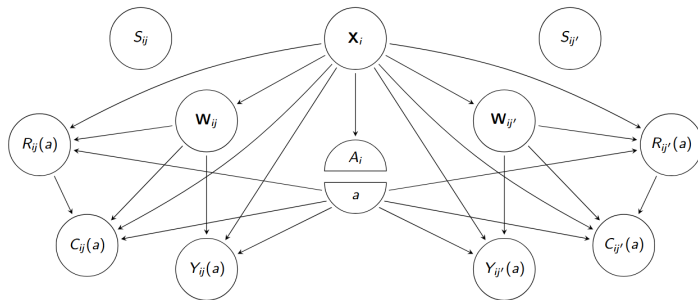


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$$\Pr(R_{ij} = 1, C_{ij} = 0 | \mathbf{X}_i = \mathbf{x}, A_i = a, \mathbf{W}_{ij} = \mathbf{w}) > 0 \quad \forall \mathbf{x}, \mathbf{w} \text{ with } f_{\mathbf{X}, \mathbf{W}}(\mathbf{x}, \mathbf{w}) > 0$$



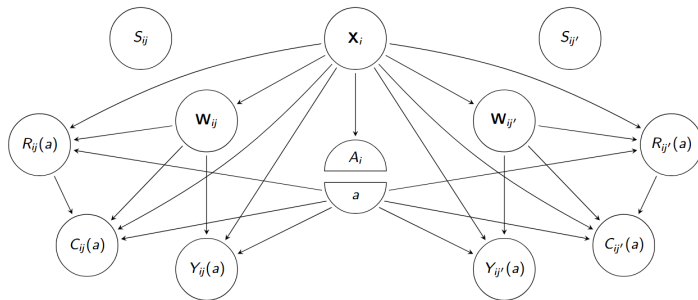
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- ▶ **Conditional independence of potential outcome and response:** $Y_{ij}(a) \perp\!\!\!\perp R_{ij}(a) | A_i, \mathbf{X}_i, \mathbf{W}_{ij}$



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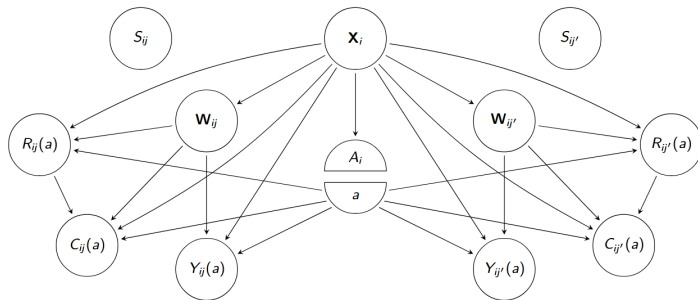
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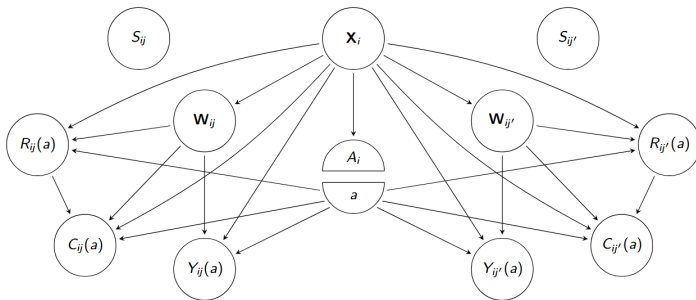
- **Conditional independence of potential outcome and censoring among responders:**

$$Y_{ij}(a) \perp\!\!\!\perp C_{ij}(a) | A_i, \mathbf{X}_i, \mathbf{W}_{ij}, R_{ij} = 1$$



Identification

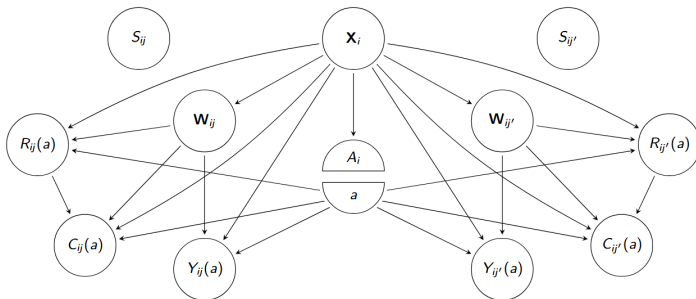
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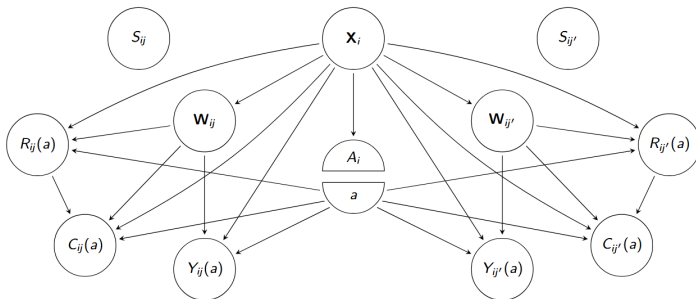
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- Inverse Probability Weighting:

$$\eta(a) = E\left\{ \frac{\mathbb{1}(A_i = a, S_{ij} = 1, R_{ij} = 1, C_{ij} = 0) Y_{ij}}{\Pr(A_i = a, S_{ij} = 1, R_{ij} = 1, C_{ij} = 0 | \mathbf{X}_i, \mathbf{W}_{ij})} \right\}$$



Estimation: G-Formula

- 1 Fit the **outcome regression model** among **uncensored responders**

$$\mu_a(\mathbf{x}, \mathbf{w}) = E(Y_{ij} | R_{ij} = 1, C_{ij} = 0, A_i = a, \mathbf{X}_i = \mathbf{x}, \mathbf{W}_{ij} = \mathbf{w})$$

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- 1 Fit the **outcome regression model** among **uncensored responders**

$$\mu_a(\mathbf{x}, \mathbf{w}) = \underbrace{E(Y_{ij})}_{\text{expected outcome}} \mid R_{ij} = 1, C_{ij} = 0, A_i = a, \mathbf{X}_i = \mathbf{x}, \mathbf{W}_{ij} = \mathbf{w})$$

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- 2 Marginalize the estimator $\hat{\mu}_a(\mathbf{X}, \mathbf{W})$ over the distribution of covariates in the **auxiliary data**:

$$\hat{\eta}^{GF}(a) = \underbrace{\frac{1}{n_{\text{aux}}} \sum_{i=1}^m \sum_{j=1}^{n_i} \mathbb{1}(S_{ij} = 0) \hat{\mu}_a(\mathbf{X}_i, \mathbf{W}_{ij})}_{\text{sample mean in auxiliary data}}$$

Estimation: Inverse Probability Weighting

- 1 Fit the **joint propensity score model**

$$\pi_a(\mathbf{x}, \mathbf{w}) = \Pr(A_i = a, R_{ij} = 1, C_{ij} = 0 | \mathbf{X}_i = \mathbf{x}, \mathbf{W}_{ij} = \mathbf{w})$$

Estimation: Inverse Probability Weighting

- 1 Fit the **joint propensity score model**

$$\pi_a(\mathbf{x}, \mathbf{w}) = \Pr \left(\underbrace{(A_i = a, R_{ij} = 1, C_{ij} = 0)}_{\text{treatment } a \text{ and uncensored response}} \mid \mathbf{X}_i = \mathbf{x}, \mathbf{W}_{ij} = \mathbf{w} \right)$$

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- 2 Take the **weighted mean outcome**

$$\hat{\eta}^{IPW}(a) = \frac{1}{n^{\text{trial}}} \sum_{i=1}^m \sum_{j=1}^{n_i} \frac{\mathbb{1}(A_i = a, S_{ij} = 1, R_{ij} = 1, C_{ij} = 0) Y_{ij}}{\hat{\pi}_a(\mathbf{X}_i, \mathbf{W}_{ij})}$$

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- 2 Take the **weighted mean outcome**

$$\hat{\eta}^{IPW}(a) = \underbrace{\frac{1}{n^{\text{trial}}}}_{\text{trial size}} \sum_{i=1}^m \sum_{j=1}^{n_i} \frac{\overbrace{\mathbb{1}(A_i = a, S_{ij} = 1, R_{ij} = 1, C_{ij} = 0) Y_{ij}}^{\text{outcomes among uncensored responders with treatment } a}}{\underbrace{\hat{\pi}_a(\mathbf{X}_i, \mathbf{W}_{ij})}_{\text{inverse probability weights}}}$$

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- 2 Take the **weighted mean outcome**

$$\hat{\eta}^{IPW}(a) = \underbrace{\frac{1}{\hat{n}^{\text{trial}}}}_{\text{Hájek estimator}} \sum_{i=1}^m \sum_{j=1}^{n_i} \frac{\overbrace{\mathbb{1}(A_i = a, S_{ij} = 1, R_{ij} = 1, C_{ij} = 0) Y_{ij}}^{\text{outcomes among uncensored responders with treatment } a}}{\underbrace{\hat{\pi}_a(\mathbf{X}_i, \mathbf{W}_{ij})}_{\text{inverse probability weights}}}$$

Joint Propensity Scores

$$\pi_a(\mathbf{x}, \mathbf{w}) = \Pr(A_i = a, R_{ij} = 1, C_{ij} = 0 | \mathbf{X}_i = \mathbf{x}, \mathbf{W}_{ij} = \mathbf{w})$$

Joint Propensity Scores

$$\begin{aligned}\pi_a(\mathbf{x}, \mathbf{w}) &= \Pr(A_i = a, R_{ij} = 1, C_{ij} = 0 | \mathbf{X}_i = \mathbf{x}, \mathbf{W}_{ij} = \mathbf{w}) \\ &= \Pr(A_i = a, R_{ij} = 1 | \mathbf{X}_i = \mathbf{x}, \mathbf{W}_{ij} = \mathbf{w}) \times \Pr(C_{ij} = 0 | A_i = a, R_{ij} = 1, \mathbf{X}_i = \mathbf{x}, \mathbf{W}_{ij} = \mathbf{w})\end{aligned}$$

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Response and Treatment Mechanism: $\pi_a^{AR}(\mathbf{x}, \mathbf{w})$

$$= \Pr(A_i = 1, R_{ij} = 1) \frac{f_{\mathbf{X}, \mathbf{W} | A, R}(\mathbf{x}, \mathbf{w} | a, 1)}{f_{\mathbf{X}, \mathbf{W}}(\mathbf{x}, \mathbf{w})}$$

Joint Propensity Scores

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- Estimate this ratio of densities following an approach in [Sugiyama et al. \(2010\)](#)

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Joint Propensity Scores

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Censoring Mechanism: $\pi_a^C(\mathbf{x}, \mathbf{w})$

- Estimate using responders in the trial (for whom $A_i, \mathbf{X}_i, \mathbf{W}_{ij}, C_{ij}$ are available)

Estimation: Augmented Inverse Probability Weighting

- 1 Fit the **outcome regression model** μ_a

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- 3 AIPW estimator: $\hat{\eta}^{AIPW}(a) =$

$$\sum_{i=1}^m \sum_{j=1}^{n_i} \left[\frac{\mathbb{1}(A_i = a, S_{ij} = 1, R_{ij} = 1, C_{ij} = 0)}{\hat{n}^{\text{trial}} \hat{\pi}_a(\mathbf{X}_i, \mathbf{W}_{ij})} \{Y_{ij} - \hat{\mu}_a(\mathbf{X}_i, \mathbf{W}_{ij})\} + \frac{\mathbb{1}(S_{ij} = 0)}{n^{\text{aux}}} \hat{\mu}_a(\mathbf{X}_i, \mathbf{W}_{ij}) \right]$$

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$\Rightarrow$ **Double robustness** to parametric model specifications for μ_a, π_a

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$\implies$ **Double robustness** to parametric model specifications for μ_a, π_a

Crash Course on M-Estimation

Estimating (Score) Function: a function of the observed data and the parameter of interest

$$\underbrace{\Psi}_{\text{score fun.}} \left(\quad ; \quad \right)$$

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The M-estimator $\hat{\theta}$ is the solution to

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$\hat{\theta}$ is consistent and asymptotically normal and has a simple variance estimator ([Stefanski and Boos, 2002](#))

Inference: Estimating Equations

Each of the proposed estimators can be expressed as an **M-estimator**:

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$\hat{\eta}^{GF}$ has estimating function

$$\Psi_{GF}(\mathbf{D}_{ij}; \theta_{GF}) = \begin{bmatrix} \Psi_{\mu}(\mathbf{D}_{ij}, \beta) \\ \frac{1}{p^{\text{aux}}} \mathbb{1}(S_{ij} = 0) \mu_0(\mathbf{X}_i, \mathbf{W}_{ij}; \beta) - \eta(0) \\ \frac{1}{p^{\text{aux}}} \mathbb{1}(S_{ij} = 1) \mu_1(\mathbf{X}_i, \mathbf{W}_{ij}; \beta) - \eta(1) \end{bmatrix},$$

where

- Ψ_{μ} is an estimating function for fitting the outcome model $\mu_a(\mathbf{x}, \mathbf{w}; \beta)$
- $\theta_{GF} = [\beta, \eta(0), \eta(1)]$
- $p^{\text{aux}} = n^{\text{aux}}/n$

Inference: Estimating Equations

Each of the proposed estimators can be expressed as an **M-estimator**:

$\hat{\eta}^{IPW}$ has estimating function

$$\Psi_{IPW}(\mathbf{D}_{ij}; \theta_{IPW}) = \begin{bmatrix} \Psi_{\pi}(\mathbf{D}_{ij}, \gamma) \\ \frac{1}{p^{\text{trial}}} \frac{\mathbb{1}(A_i=0, S_{ij}=1, R_{ij}=1, C_{ij}=0) Y_{ij}}{\hat{\pi}_0(\mathbf{x}_i, \mathbf{w}_{ij}; \gamma)} - \eta(0) \\ \frac{1}{p^{\text{trial}}} \frac{\mathbb{1}(A_i=1, S_{ij}=1, R_{ij}=1, C_{ij}=0) Y_{ij}}{\hat{\pi}_1(\mathbf{x}_i, \mathbf{w}_{ij}; \gamma)} - \eta(1) \end{bmatrix},$$

where

- Ψ_{π} is an estimating function for fitting the joint propensity score model $\pi_a(\mathbf{x}, \mathbf{w}; \gamma)$
- $\theta_{IPW} = [\gamma, \eta(0), \eta(1)]$
- $p^{\text{trial}} = n^{\text{trial}}/n$

Inference: Estimating Equations

Each of the proposed estimators can be expressed as an **M-estimator**:

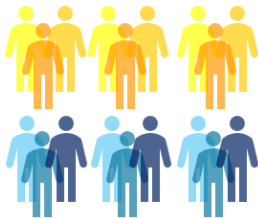
$\hat{\eta}^{AIPW}$ has estimating function

$$\Psi_{AIPW}(\mathbf{D}_{ij}; \theta_{AIPW}) = \begin{bmatrix} \Psi_{\mu}(\mathbf{D}_{ij}, \beta) \\ \Psi_{\pi}(\mathbf{D}_{ij}, \gamma) \\ \frac{\mathbb{1}(A_i=0, S_{ij}=1, R_{ij}=1, C_{ij}=0)}{p^{\text{trial}} \pi_0(\mathbf{X}_i, \mathbf{W}_{ij}; \gamma)} \{Y_{ij} - \mu_0(\mathbf{X}_i, \mathbf{W}_{ij}; \beta)\} + \frac{\mathbb{1}(S_{ij}=0)}{p^{\text{aux}}} \mu_0(\mathbf{X}_i, \mathbf{W}_{ij}; \beta) - \eta(0) \\ \frac{\mathbb{1}(A_i=1, S_{ij}=1, R_{ij}=1, C_{ij}=0)}{p^{\text{trial}} \pi_1(\mathbf{X}_i, \mathbf{W}_{ij}; \gamma)} \{Y_{ij} - \mu_1(\mathbf{X}_i, \mathbf{W}_{ij}; \beta)\} + \frac{\mathbb{1}(S_{ij}=0)}{p^{\text{aux}}} \mu_1(\mathbf{X}_i, \mathbf{W}_{ij}; \beta) - \eta(1) \end{bmatrix},$$

where

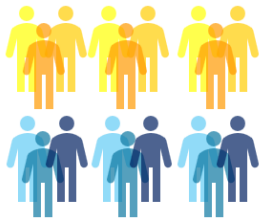
- $\theta_{AIPW} = [\beta, \gamma, \eta(0), \eta(1)]$

Simulation Study



Simulation Study

$$X \sim N(0,1)$$

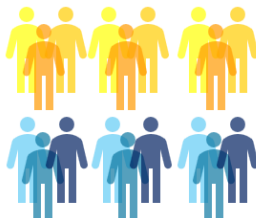


Simulation Study

$$X \sim N(0,1)$$

$$W_1 \sim \text{Bernoulli}(0.5)$$

$$W_2 \sim N(0,1)$$

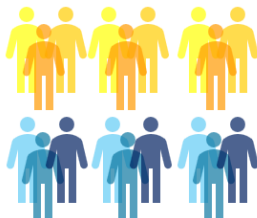


Simulation Study

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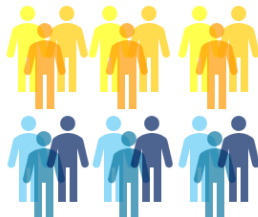
Treatment marginally protective,
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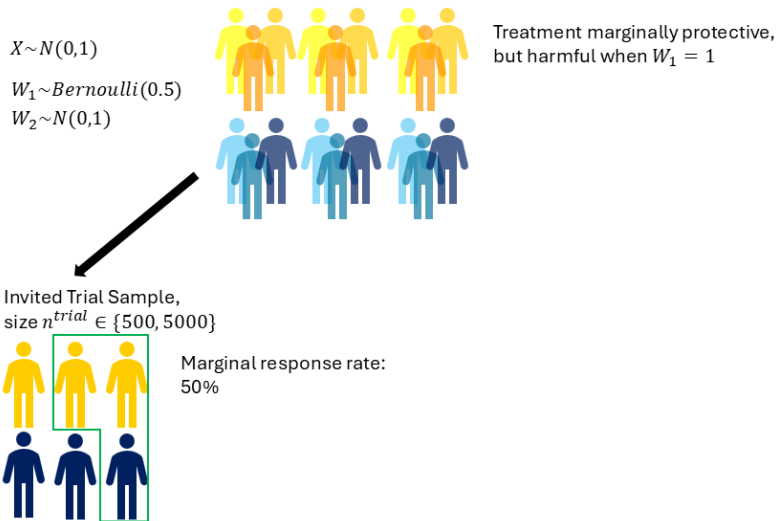


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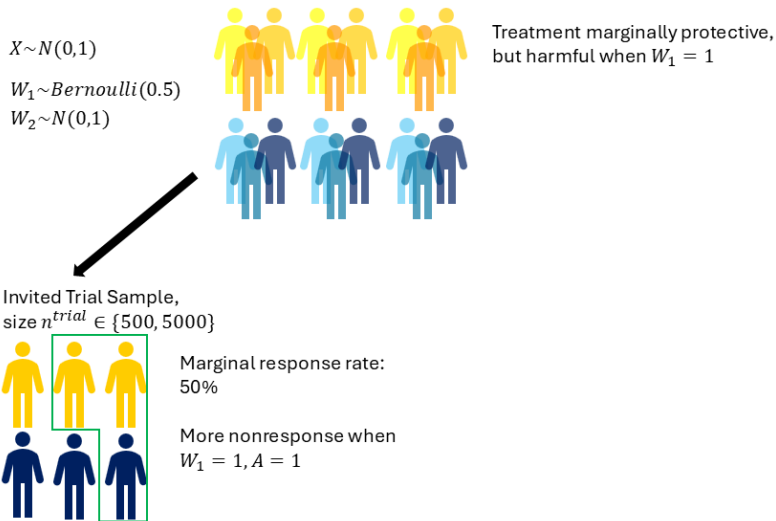
Invited Trial Sample,
size $n^{trial} \in \{500, 5000\}$



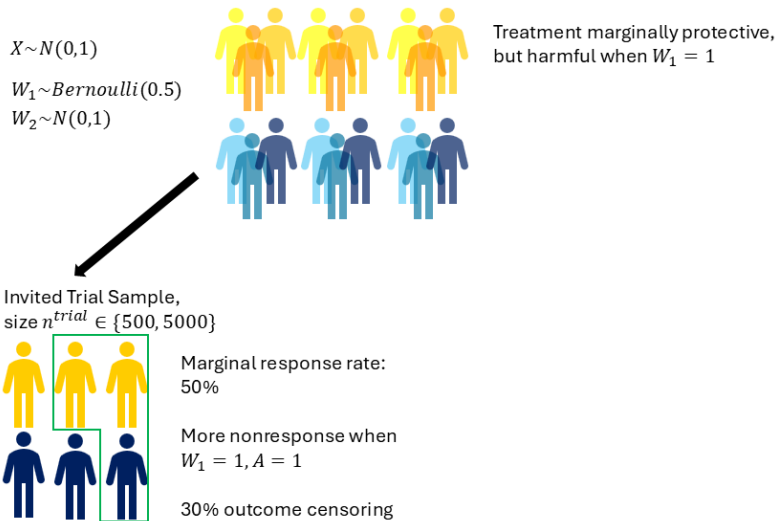
Simulation Study



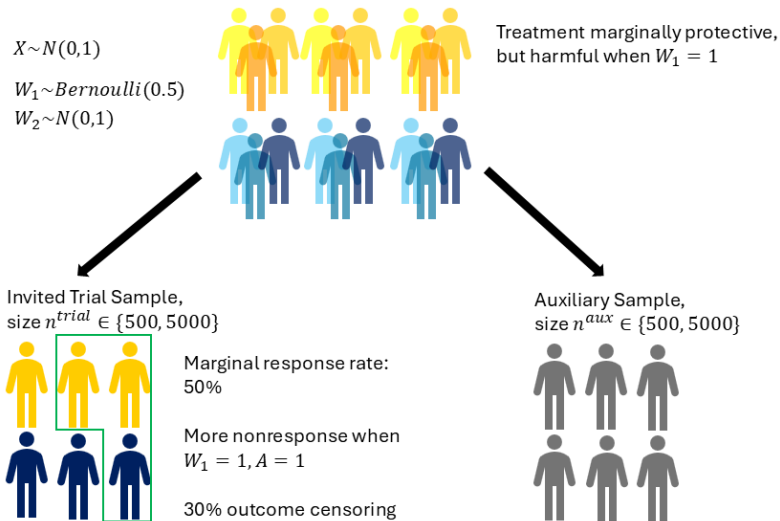
Simulation Study



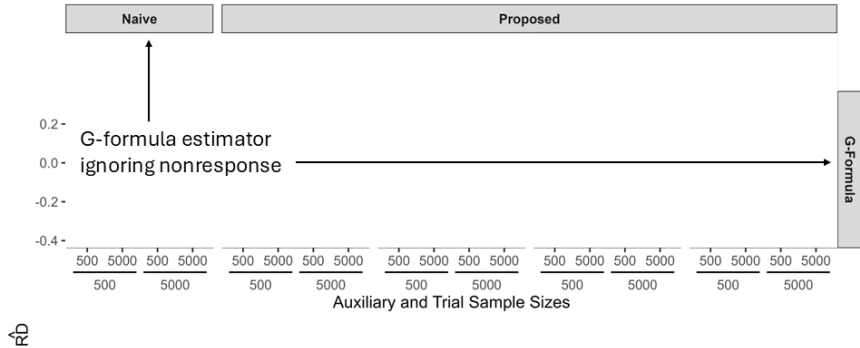
Simulation Study



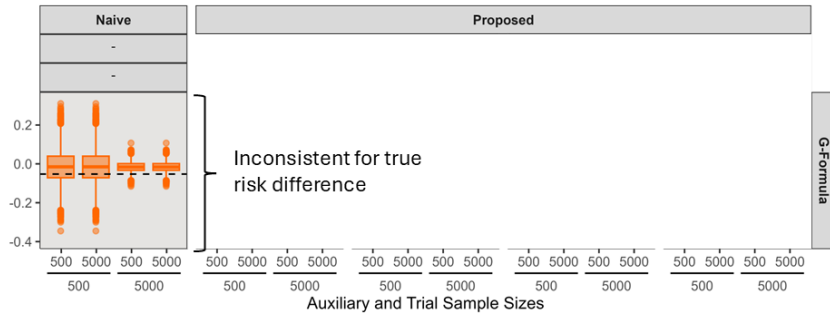
Simulation Study



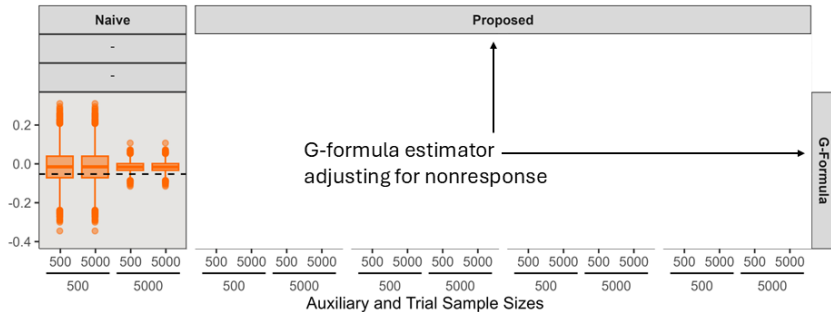




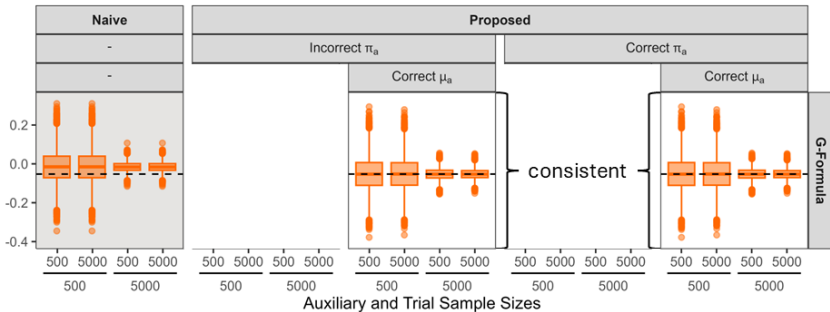
$\hat{RD}$



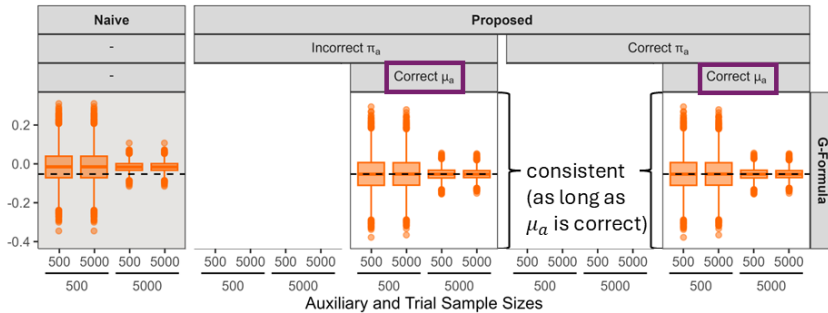
RD



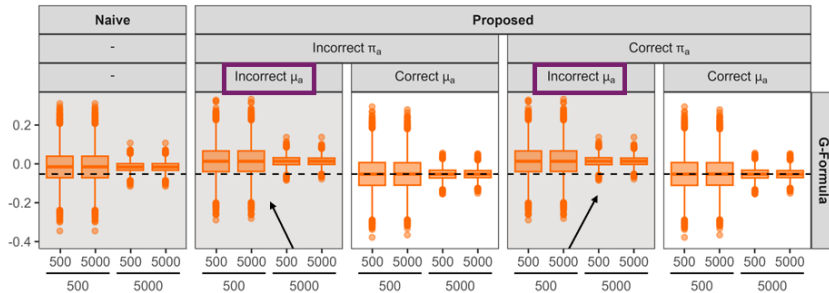
RD



RD

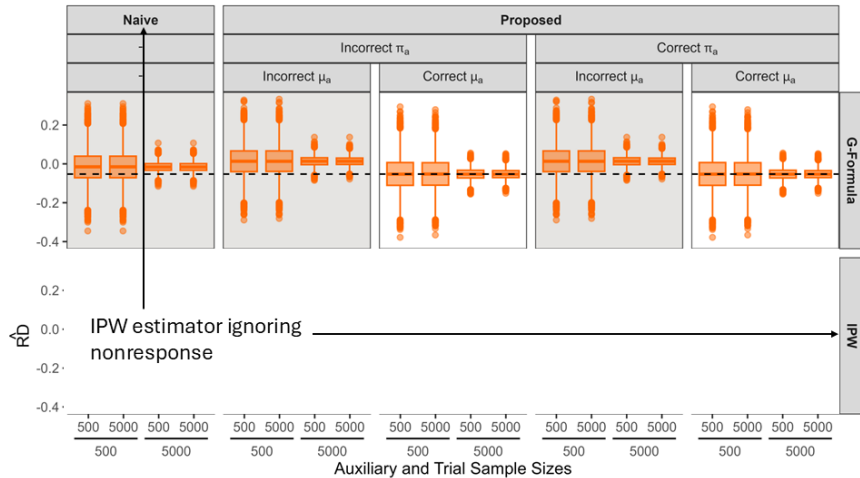


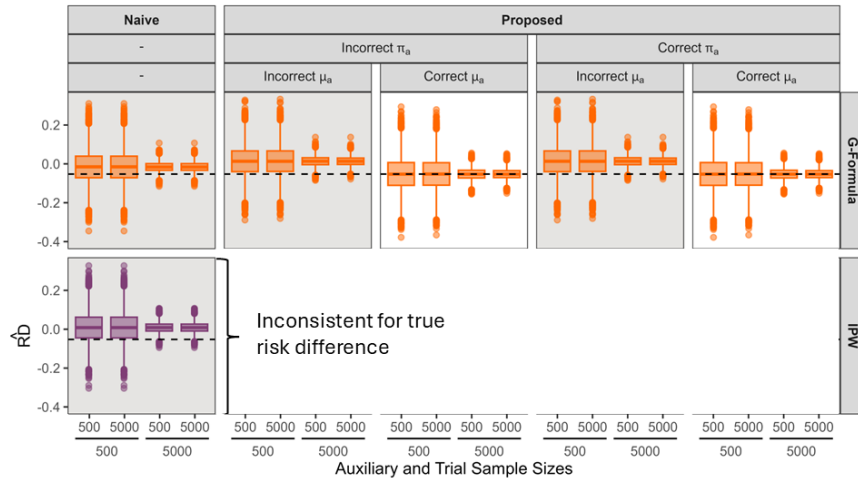
RB

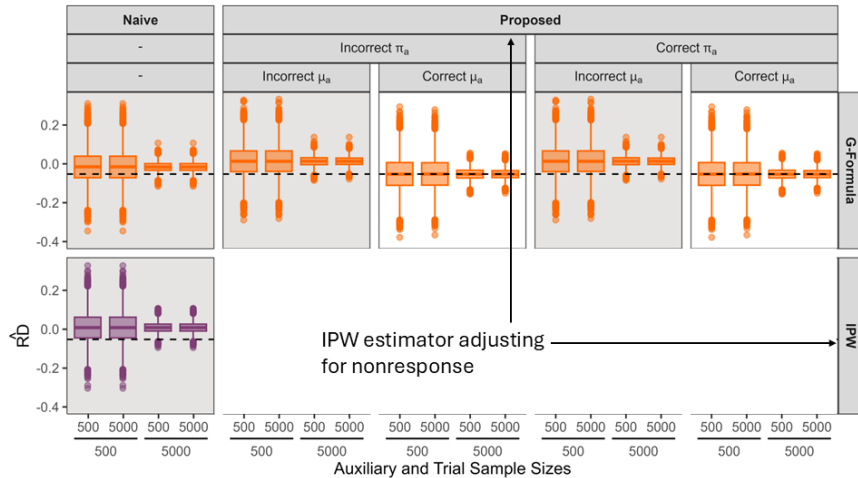


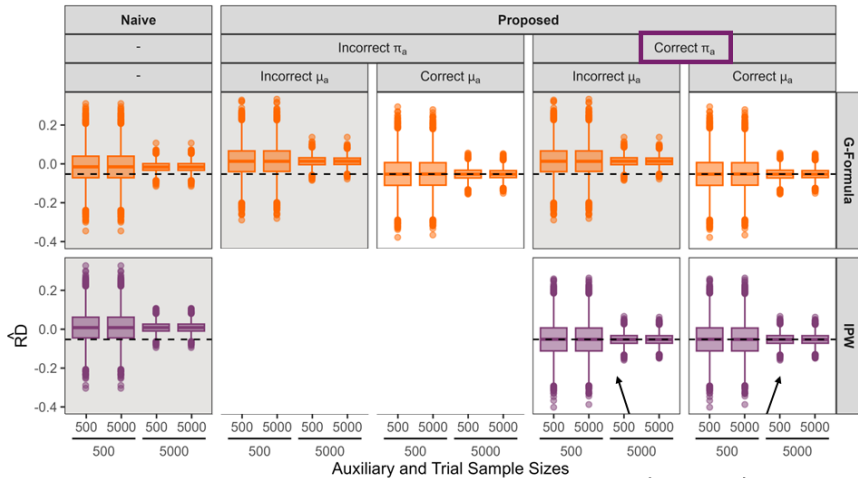
Auxiliary and Trial Sample Sizes

Inconsistent if μ_a is incorrect

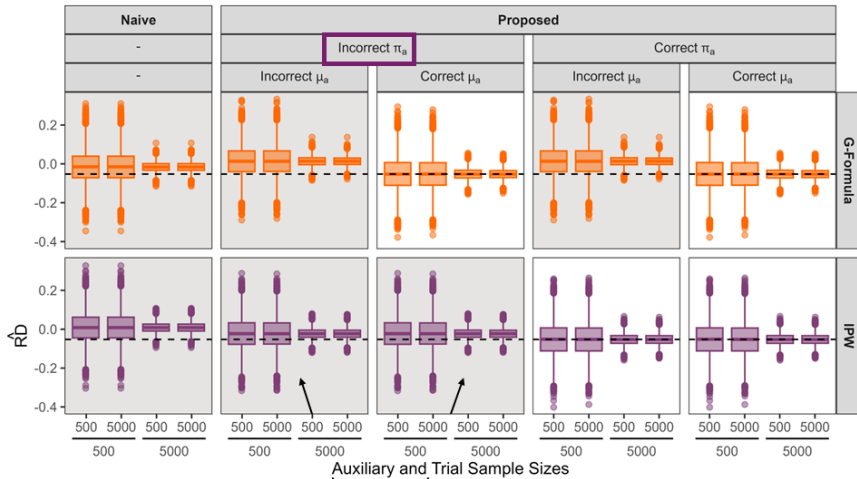




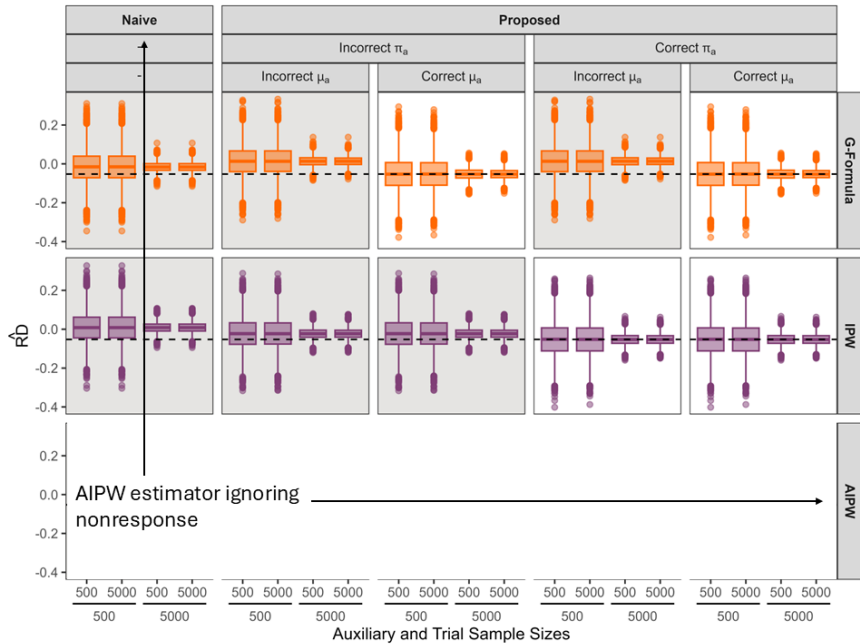


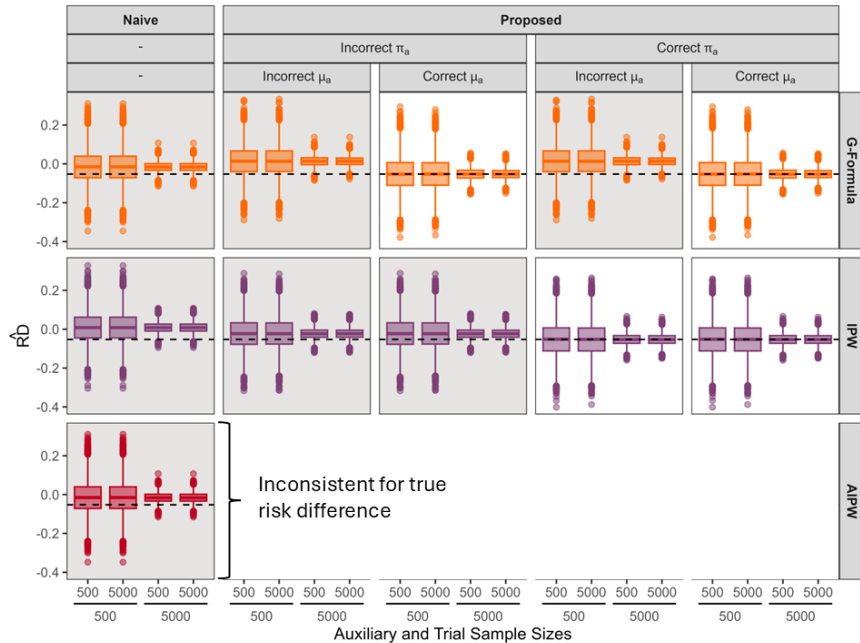


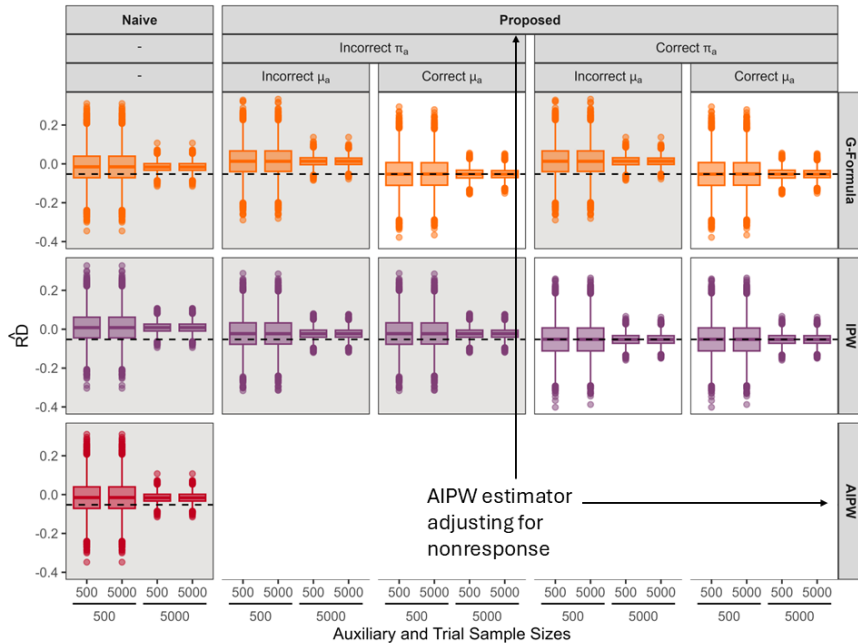
consistent if π_a
is correct

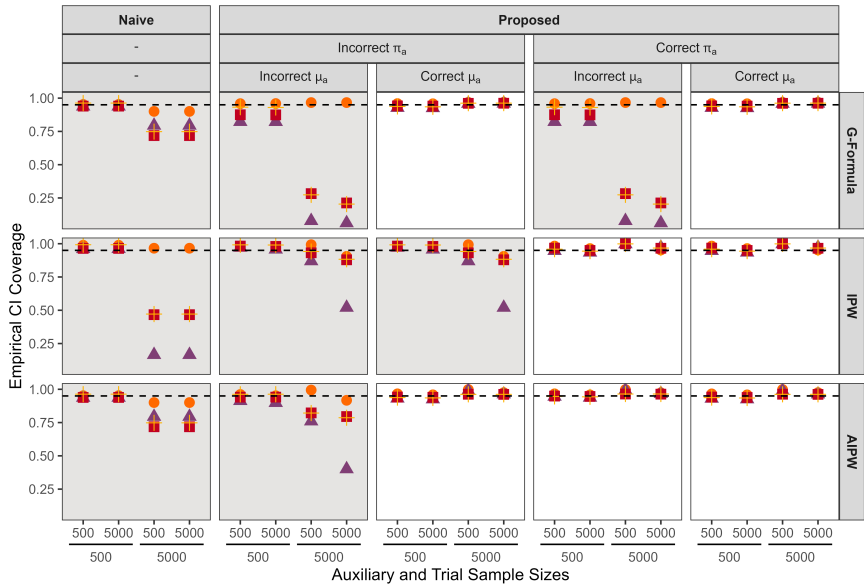


inconsistent if
 π_a is incorrect









Estimand $\eta(0)$ $\eta(1)$ RD RR

PopART Application

- Trial data: PopART trial ([Hayes et al., 2014](#), [2019](#))

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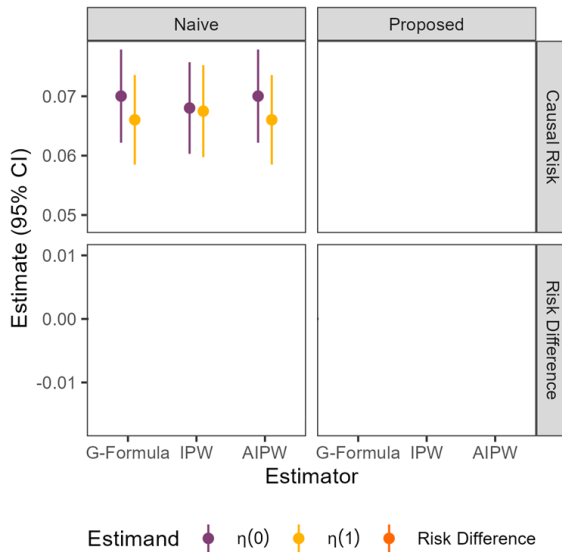
	PopART Trial Responders				Auxiliary Data
	Overall <i>N</i> = 29,130	A <i>N</i> = 9,594	B <i>N</i> = 10,235	C (Control) <i>N</i> = 9,301	<i>N</i> = 5,076
Sex and Male Circumcision					
Female	19,494 (68%)	6,547 (69%)	6,828 (68%)	6,119 (68%)	54%
Male (Circumcised)	4,190 (15%)	1,403 (15%)	1,516 (15%)	1,271 (14%)	22%
Male (Uncircumcised)	4,847 (17%)	1,474 (16%)	1,725 (17%)	1,648 (18%)	24%

PopART Application

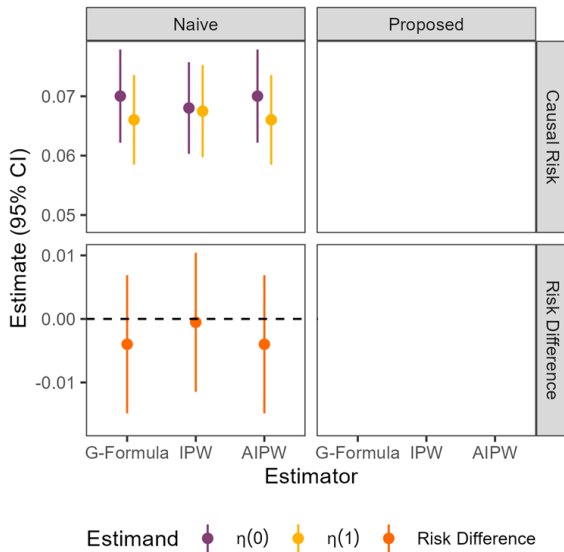
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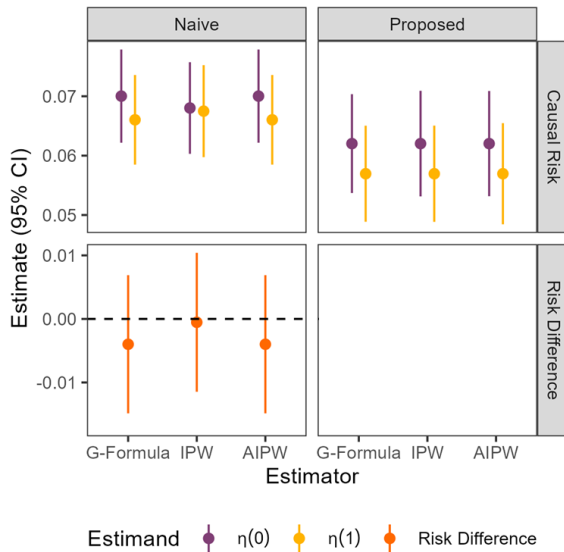
PopART Results



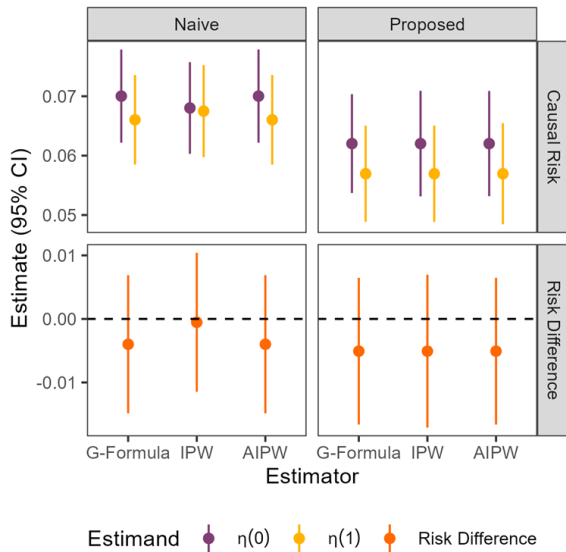
PopART Results



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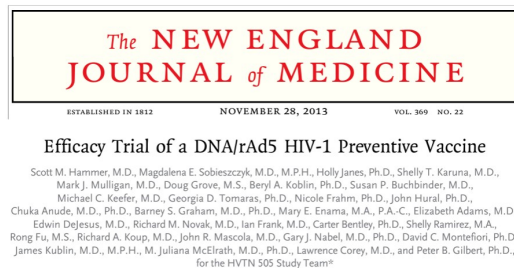
Summary of Project 3

- Developed estimators to adjust for differential nonresponse and outcome censoring in cluster RCT
 - ▶ Established theoretical properties (consistency, asymptotic normality, double robustness of AIPW)
 - ▶ Demonstrated empirical performance through simulations
- Applied the method to the **PopART trial**, along with auxiliary data from national surveys
- **Future Directions:**
 - ▶ show semiparametric efficiency of AIPW estimator
 - ▶ derive asymptotic results for AIPW estimator given nonparametric estimators of μ_a, π_a
- **Status:** in preparation to submit to *Annals of Applied Statistics*

Project 1 : Addressing Confounding and Continuous Exposure Measurement Error Using Corrected Score Functions

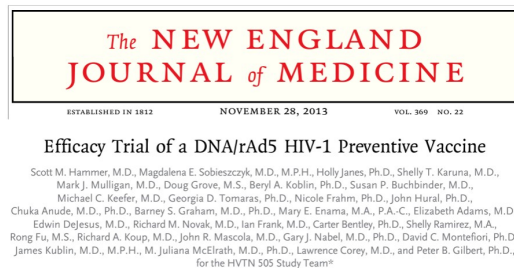
Brian Richardson, Brian Blette, Peter Gilbert, Michael Hudgens (2025)

Motivation: HVTN 505 trial



- **HVTN 505 trial:** trial of a preventive HIV vaccine

Motivation: HVTN 505 trial



- **HVTN 505 trial:** trial of a preventive HIV vaccine
- Stopped early after reaching predetermined cutoffs for efficacy futility ([Hammer et al., 2013](#))

Motivation: HVTN 505 trial

The Journal of Infectious Diseases

MAJOR ARTICLE



Higher T-Cell Responses Induced by DNA/rAd5 HIV-1 Preventive Vaccine Are Associated With Lower HIV-1 Infection Risk in an Efficacy Trial

Holly E. Janes,¹ Kristen W. Cohen,² Nicole Frahm,³ Stephen C. De Rosa,¹ Brittany Sanchez,¹ John Hural,¹ Craig A. Magaret,¹ Shelly Karuna,¹ Carter Bentley,¹ Raphael Gottardo,¹ Greg Finak,¹ Douglas Grove,² Mingchao Shen,¹ Barney S. Graham,³ Richard A. Koup,³ Mark J. Mulligan,⁴ Beryl Koblin,⁵ Susan P. Buchbinder,⁶ Michael C. Keefer,⁷ Elizabeth Adams,⁸ Chuka Anude,^{8a} Lawrence Corey,¹ Magdalena Sobieszczyk,^{1b} Scott M. Hammer,^{1b} Peter B. Gilbert,¹ and M. Juliana McElrath¹

- Several biomarkers correlated with HIV among vaccine recipients ([Janes et al., 2017](#); [Fong et al., 2018](#); [Neidich et al., 2019](#))

Motivation: HVTN 505 trial

The Journal of Infectious Diseases

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- Is there a **causal** relationship between these biomarkers and HIV?

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The Journal of Infectious Diseases

MAJOR ARTICLE

 IDSA
Infectious Diseases Society of America

 hivma
HIV Medicine Association

 OXFORD

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The Journal of Infectious Diseases

MAJOR ARTICLE



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- Is there a **causal** relationship between these biomarkers and HIV?
- Biomarker-HIV relationship is **confounded**
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To estimate the **causal effect** of a continuous exposure on an outcome when

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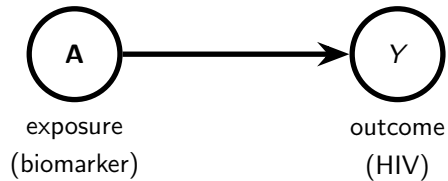
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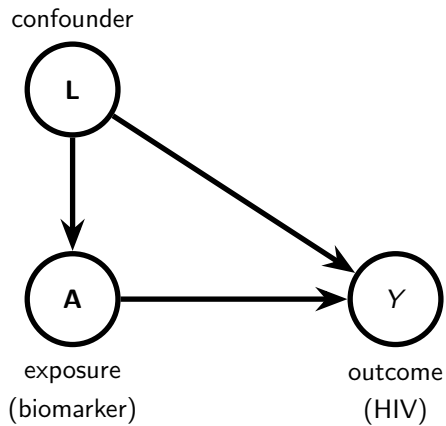
To estimate the **causal effect** of a continuous exposure on an outcome when

- (i) The exposure-outcome relationship is potentially **confounded**
- (ii) The exposure is **measured with error**

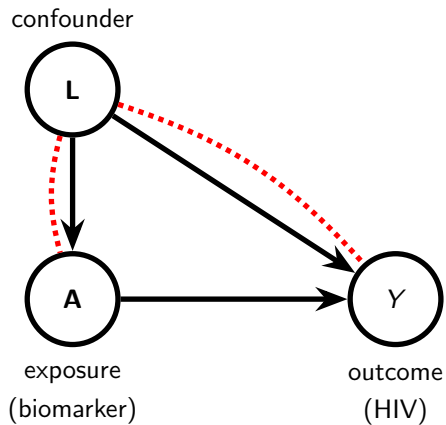
Confounding



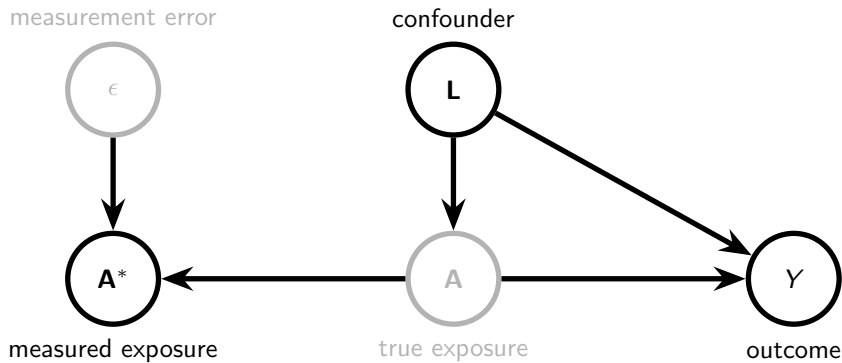
Confounding



Confounding



Measurement Error



Addressing Confounding Alone

- G-Formula
- IPW
- Doubly Robust Estimator

Addressing Confounding Alone

- G-Formula

$$\Psi_{0-GF}(Y, \mathbf{L}, \mathbf{A}; \theta_{GF}) = \begin{bmatrix} \{Y - \mu(\mathbf{L}, \mathbf{A}; \beta)\} \partial_{\beta} \mu(\mathbf{L}, \mathbf{A}; \beta) \\ \eta(\mathbf{a}) - \mu(\mathbf{L}, \mathbf{a}; \beta) \end{bmatrix}$$

- IPW
- Doubly Robust Estimator

Addressing Confounding Alone

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- IPW

$$\Psi_{0-IPW}(Y, \mathbf{L}, \mathbf{A}; \theta_{IPW}) = \left[\begin{array}{c} \Psi_{PS}(\mathbf{L}, \mathbf{A}) \\ SW(\mathbf{L}, \mathbf{A}) \{Y - \eta(\mathbf{A}; \gamma)\} \partial_{\gamma} \eta(\mathbf{A}; \gamma) \end{array} \right]$$

- Doubly Robust Estimator

Addressing Confounding Alone

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$$\psi_{0-GF}(Y, \mathbf{L}, \mathbf{A}; \theta_{GF}) = \left[\begin{array}{c} \{Y - \mu(\mathbf{L}, \mathbf{A}; \beta)\} \partial_{\beta} \mu(\mathbf{L}, \mathbf{A}; \beta) \\ \eta(\mathbf{a}) - \mu(\mathbf{L}, \mathbf{a}; \beta) \end{array} \right]$$

- IPW

$$\psi_{0-IPW}(Y, \mathbf{L}, \mathbf{A}; \theta_{IPW}) = \left[\begin{array}{c} \Psi_{PS}(\mathbf{L}, \mathbf{A}) \\ SW(\mathbf{L}, \mathbf{A}) \{Y - \eta(\mathbf{A}; \gamma)\} \partial_{\gamma} \eta(\mathbf{A}; \gamma) \end{array} \right]$$

- Doubly Robust Estimator

$$\psi_{0-DR}(Y, \mathbf{L}, \mathbf{A}; \theta_{DR}) = \left[\begin{array}{c} \Psi_{PS}(\mathbf{L}, \mathbf{A}) \\ SW(\mathbf{L}, \mathbf{A}) \{Y - \mu(\mathbf{L}, \mathbf{A}; \beta)\} \partial_{\beta} \mu(\mathbf{L}, \mathbf{A}; \beta) \\ \eta(\mathbf{a}) - \mu(\mathbf{L}, \mathbf{a}; \beta) \end{array} \right]$$

Addressing Confounding and Measurement Error

Can we just substitute $\mathbf{A}^*$ for $\mathbf{A}$ and find the solution to

$$\sum_{i=1}^n \psi_0(Y_i, \mathbf{L}_i, \underbrace{\mathbf{A}_i^*}_{\text{mismeasured}}; \theta) = \mathbf{0}?$$

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No! This leads to bias in $\hat{\theta}$ because

$$E\{\psi_0(Y, \mathbf{L}, \mathbf{A}^*; \theta_0)\} \neq \mathbf{0}.$$

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No! This leads to bias in $\hat{\theta}$ because

$$E\{\psi_0(Y, \mathbf{L}, \mathbf{A}^*; \theta_0)\} \neq \mathbf{0}.$$

We need a new score function ψ_{CS} such that

$$E\{\underbrace{\psi_{CS}}_{\text{new score fun.}}(Y, \mathbf{L}, \underbrace{\mathbf{A}^*}_{\text{mismeasured}}; \theta_0)\} = \mathbf{0}.$$

Corrected Score Functions

Given the “oracle” score function Ψ_0 , the “corrected score” function Ψ_{CS} can be created following Novick and Stefanski (2002) ([Novick and Stefanski, 2002](#)):

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$$\Psi_{CS}(Y, \mathbf{L}, \mathbf{A}^*; \theta) \equiv E[\text{Re}\{\Psi_0(Y, \mathbf{L}, \mathbf{A}^* + i\tilde{\epsilon}; \theta)\} | Y, \mathbf{L}, \mathbf{A}^*]$$

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The G-Formula, IPW, and DR score functions can all be corrected in this way

$$\Psi_{0-GF} \longrightarrow \Psi_{CS-GF}$$

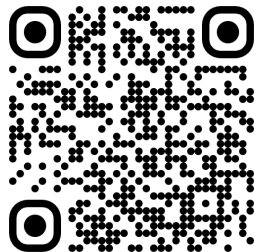
$$\Psi_{0-IPW} \longrightarrow \Psi_{CS-IPW}$$

$$\Psi_{0-DR} \longrightarrow \Psi_{CS-DR}$$

Summary of Project 1



Paper in *Biometrics* (2025)



GitHub R package

Mismex: Causal Inference for Mismeasured Exposures

Acknowledgments

- add photos

Thank you! Questions?

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Supplement to Project 2: Physical Distancing to Reduce Transmission of Influenza-Like-Illness on College Campuses: the eX-FLU Trial

Brian Richardson, Allison Aiello, Michael Hudgens

Choice of Test Statistic

Test statistics: “proportion of possible transmission events attributable to students in the intervention group:”

$$T^{**}(\mathbf{Z}, \bar{\mathbf{A}}, \bar{\mathbf{Y}}) = \frac{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j>i} Z_i E_{ijk}^{**}}{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j>i} E_{ijk}^{**}}.$$

	From Infected to Infected	From Infected
Contact at $k - 1$	$E_{ijk}^{11} = Y_i^{k-1} A_{ij}^{k-1} Y_j^k$	$E_{ijk}^{12} = Y_i^{k-1} A_{ij}^{k-1}$
Contact at k	$E_{ijk}^{21} = Y_i^{k-1} A_{ij}^k Y_j^k$	$E_{ijk}^{22} = Y_i^{k-1} A_{ij}^k$
Contact at $k - 1$ or k	$E_{ijk}^{31} = Y_i^{k-1} (A_{ij}^{k-1} \vee A_{ij}^k) Y_j^k$	$E_{ijk}^{32} = Y_i^{k-1} (A_{ij}^{k-1} \vee A_{ij}^k)$
Contact at $k - 1$ and k	$E_{ijk}^{41} = Y_i^{k-1} (A_{ij}^{k-1} * A_{ij}^k) Y_j^k$	$E_{ijk}^{42} = Y_i^{k-1} (A_{ij}^{k-1} * A_{ij}^k)$

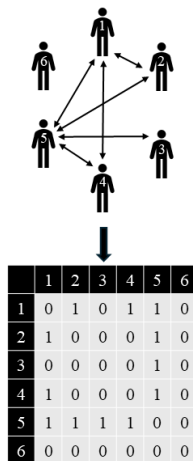
Table: Definitions of a possible transmission event E_{ijk} from student i to student j at time k . Y_i^k is an indicator for student i being infected at week k , A_{ij}^k is an indicator for students i and j being in contact at week k , and $a \vee b$ denotes the maximum of a and b .

Exponential Family Random Graph Models

Goal: model a network (**A**) given covariates (**X**)

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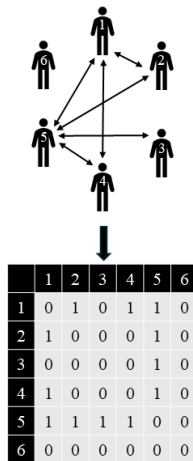


Exponential Family Random Graph Models

Goal: model a network (**A**) given covariates (**X**)

An **ERGM** assumes:

$$\Pr_{\theta}(\mathbf{A} = \mathbf{a} | \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta \cdot \mathbf{g}(\mathbf{a}, \mathbf{x})\}}{\kappa(\theta, \mathcal{A}, \mathbf{x})}$$



Exponential Family Random Graph Models

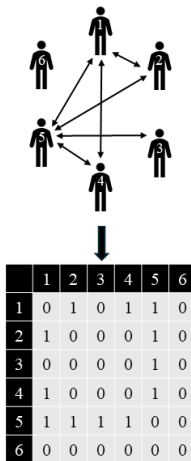
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- normalizing constant:

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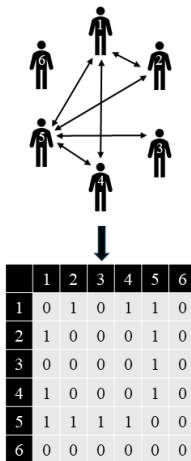
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- $\mathcal{A}$: space of possible networks



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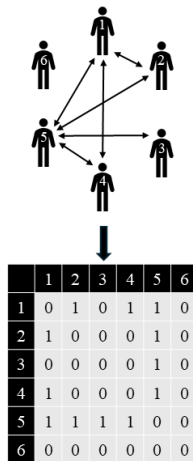
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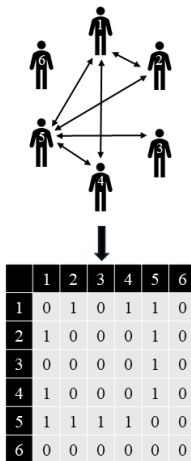
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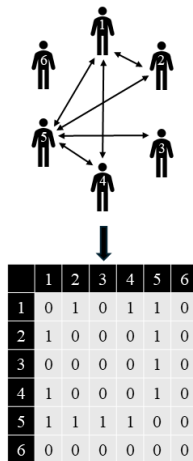
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 - ▶ number of edges



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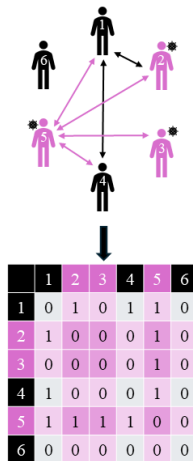
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 - ▶ number of edges
 - ▶ number of edges touching treated students



Exponential Family Random Graph Models

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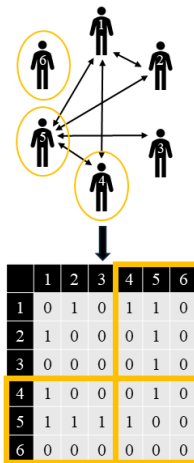
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 - ▶ number of edges
 - ▶ number of edges touching treated students
 - ▶ number of edges touching infected students



Exponential Family Random Graph Models

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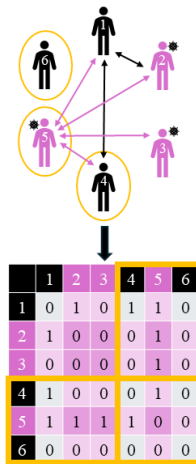
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- $\mathcal{A}$: space of possible networks
- θ : model coefficients
- $\mathbf{g}(\mathbf{a}, \mathbf{x})$: sufficient statistics
 - ▶ number of edges
 - ▶ number of edges touching treated students
 - ▶ number of edges touching infected students
 - ▶ number of edges touching treated $\times$ infected students



ERGM Model Formulation

$$\Pr_{\theta}(\mathbf{A} = \mathbf{a} | \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta \cdot \mathbf{g}(\mathbf{a}, \mathbf{x})\}}{\kappa(\theta, \mathcal{A}, \mathbf{x})},$$

$$\kappa(\theta, \mathcal{A}, \mathbf{x}) = \sum_{\mathbf{a} \in \mathcal{A}} \exp\{\theta \cdot \mathbf{g}(\mathbf{a}, \mathbf{x})\}$$

For example, in the simulation study and eX-FLU application,

$$\mathbf{g}(\mathbf{a}, \mathbf{x}) = \begin{bmatrix} \# \text{ edges} \\ \# \text{ edges touching a treated node} \\ \# \text{ edges touching an infected node} \\ \# \text{ edges touching a treated and infected node} \\ \# \text{ edges between roommate pairs} \end{bmatrix} = \begin{bmatrix} \sum_{i,j} a_{ij} \\ \sum_{i,j} a_{ij}(z_i + z_j) \\ \sum_{i,j} a_{ij}(y_i + y_j) \\ \sum_{i,j} a_{ij}(z_i y_i + z_j y_j) \\ \sum_{i,j} a_{ij} \mathbb{1}(i, j \text{ roommates}) \end{bmatrix}$$

ERGM Change Statistic Model Formulation

- **Change statistic:** $\delta_{\mathbf{g}}(\mathbf{a}, \mathbf{x})_{ij} = \mathbf{g}(\mathbf{a}_{ij}^+, \mathbf{x}) - \mathbf{g}(\mathbf{a}_{ij}^-, \mathbf{x})$ is the change in network statistic that would occur if a_{ij} were changed from 0 to 1
 - ▶ where $\mathbf{a}_{ij}^+$ and $\mathbf{a}_{ij}^-$ represent the network $\mathbf{a}$ with dyad a_{ij} set to 1 or 0, respectively
- Then the equivalent ERGM specification is

$$\text{logit}\{\Pr_{\theta}(A_{ij} = 1 | \mathbf{A}_{ij}^C = \mathbf{a}_{ij}^C, \mathbf{X} = \mathbf{x})\} = \theta^k \delta_{\mathbf{g}}(\mathbf{a}, \mathbf{x})_{ij}$$

- ▶ where $\mathbf{A}_{ij}^C$ represents all dyads in $\mathbf{A}$ except A_{ij}
- **Interpretation of θ :** the change in conditional log-odds of the network associated with a one-unit increase in the corresponding component of $\mathbf{g}(\mathbf{a}, \mathbf{x})$ resulting from switching a particular dyad A_{ij} from 0 to 1 and leaving the rest of the network fixed at $\mathbf{A}_{ij}^C$

Dyadic Independence EGRMs

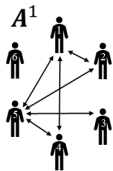
- **Dyadic independence term:** a component g of $\mathbf{g}$ in an ERGM for which the corresponding change statistic $\delta_g(\mathbf{a}, \mathbf{x})_{ij}$ can be calculated for any i, j without knowing $\mathbf{a}$
 - ▶ For example, if $g(\mathbf{a}, \mathbf{x}) = \sum_{i,j} a_{ij}(Z_i + Z_j)$ counts the number of edges touching treated nodes, then $\delta_g(\mathbf{a}, \mathbf{x})_{ij} = z_i + z_j$ doesn't depend on $\mathbf{a}$
- **Dyadic independence ERGM:** an ERGM with only dyadic independence terms
 - ▶ replace $\delta_g(\mathbf{a}, \mathbf{x})_{ij}$ with $\delta_g(\mathbf{x})_{ij}$ and write the model as

$$\text{logit}\{\Pr(A_{ij} = 1 | \mathbf{X} = \mathbf{x})\} = \theta \cdot \delta_g(\mathbf{x})_{ij}$$

- **Interpretation of θ :** the change in log-odds of the network associated with a one-unit increase in the corresponding component of $\mathbf{g}(\mathbf{a}, \mathbf{x})$ resulting from switching a particular dyad A_{ij} from 0 to 1

Separable Temporal Exponential Family Random Graph Models

A **STERGM** assumes that, at each time step $k = 1, \dots, \tau - 1$:

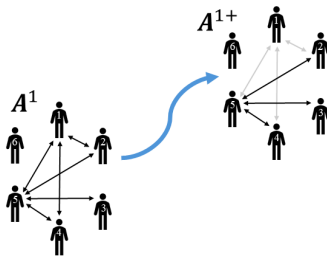


Separable Temporal Exponential Family Random Graph Models

A **STERGM** assumes that, at each time step $k = 1, \dots, \tau - 1$:

- New edges form according to a **formation** ERGM

$$\Pr_{\theta^+}(\mathbf{A}^{k+1} = \mathbf{a}^{k+1} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^+ \cdot \mathbf{g}^+(\mathbf{a}^{k+1}, \mathbf{x})\}}{\kappa\{\theta^+, \mathcal{A}^+(\mathbf{a}^k), \mathbf{x}\}}$$

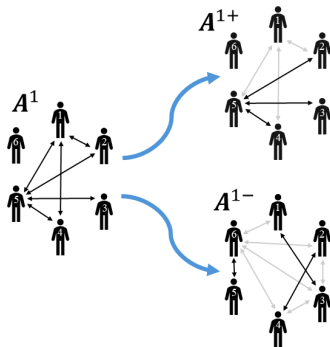


Separable Temporal Exponential Family Random Graph Models

A **STERGM** assumes that, at each time step $k = 1, \dots, \tau - 1$:

- New edges form according to a **formation** ERGM
- Old edges persist according to a **persistence** ERGM

$$\Pr(\mathbf{A}^{k-} = \mathbf{a}^{k-} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^- \cdot \mathbf{g}^-(\mathbf{a}^{k-}, \mathbf{x})\}}{\kappa\{\theta^-, \mathcal{A}^-(\mathbf{a}^k), \mathbf{x}\}}.$$

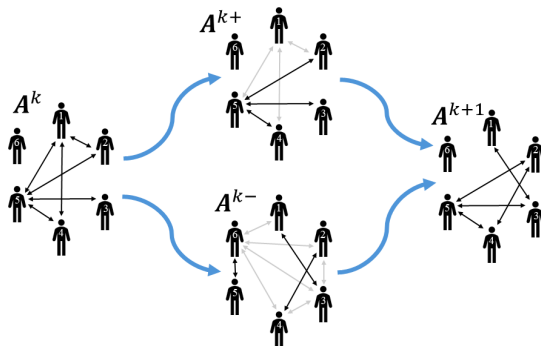


Separable Temporal Exponential Family Random Graph Models

A **STERGM** assumes that, at each time step $k = 1, \dots, \tau - 1$:

- New edges form according to a **formation** ERGM
- Old edges persist according to a **persistence** ERGM
- The network at time step $k + 1$ is the result of formation and persistence

$$\mathbf{A}^{k+1} = \underbrace{\mathbf{A}^k}_{\text{previous network}} \cup \underbrace{(\mathbf{A}^{k+} - \mathbf{A}^k)}_{\text{new edges formed}} - \underbrace{(\mathbf{A}^k - \mathbf{A}^{k-})}_{\text{old edges not persisting}}$$



STERGM Model Formulation

- **Formation model** is an ERGM conditional on only adding edges:

$$\Pr_{\theta^+}(\mathbf{A}^{k+1} = \mathbf{a}^{k+1} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^+ \cdot \mathbf{g}^+(\mathbf{a}^{k+1}, \mathbf{x})\}}{\kappa\{\theta^+, \mathcal{A}^+(\mathbf{a}^k), \mathbf{x}\}}$$

► $\mathcal{A}^+(\mathbf{a})$: space of possible networks that can be formed by adding edges to $\mathbf{a}$

- **Persistence model** is an ERGM conditional on only removing edges:

$$\Pr_{\theta^-}(\mathbf{A}^{k+1} = \mathbf{a}^{k+1} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^- \cdot \mathbf{g}^-(\mathbf{a}^{k+1}, \mathbf{x})\}}{\kappa\{\theta^-, \mathcal{A}^-(\mathbf{a}^k), \mathbf{x}\}}$$

► $\mathcal{A}^-(\mathbf{a})$: space of possible networks that can be formed by removing edges to $\mathbf{a}$

- A **STERGM** assumes the network at time $k + 1$ is then the result of applying the changes in $\mathbf{A}^{k+}$ and $\mathbf{A}^{k-}$ to $\mathbf{A}^k$:

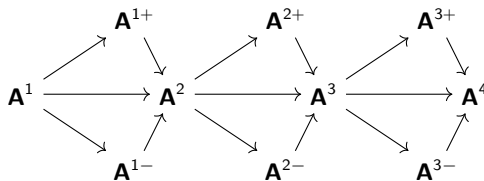
$$\mathbf{A}^{k+1} = \underbrace{\mathbf{A}^k}_{\text{previous network}} \cup \underbrace{(\mathbf{A}^{k+} - \mathbf{A}^k)}_{\text{new edges formed}} - \underbrace{(\mathbf{A}^k - \mathbf{A}^{k-})}_{\text{old edges not persisting}}$$

Separability of STERGMs

- 1 $\mathbf{A}^{k+} \perp\!\!\!\perp \mathbf{A}^{k-} | \mathbf{A}^k$, i.e., the formation and persistence processes are conditionally independent given the network at time k

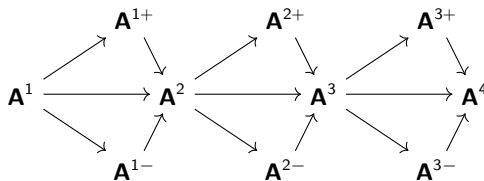
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Separability of STERGMs

- 1 $\mathbf{A}^{k+} \perp\!\!\!\perp \mathbf{A}^{k-} | \mathbf{A}^k$, i.e., the formation and persistence processes are conditionally independent given the network at time k
- 2 the parameter space for $\theta = (\theta^+, \theta^-)$ is the product of the parameter spaces for θ^+ and θ^-



Simulation Setup

- $n \in \{24, 48\}$ students were equally divided into two residence halls, each with six clusters of equal size
- Within each residence hall group, the clusters were randomized using a 50:50 allocation to either the intervention group or the control group
- Five pairs of students were randomly selected to be roommates, meaning they had contact with each other each week
- Baseline ($k = 1$) face-to-face contacts were simulated between each pair of students with probability 0.5
- Baseline infection statuses were simulated for each student with probability 0.5
- Social networks were simulated over the remaining $\tau \in \{5, 10\}$ weeks according to a STERGM with both formation and persistence models including edge count, intervention assignment Z , infection status Y , $Z \times Y$ interaction, and an offset to force constant edges between roommates
- Infection statuses were simulated for each student i at each week $k \in \{2, \dots, \tau\}$ with probability $\Pr(Y_i^k = 1 | \bar{\mathbf{A}}^{k-1}, \bar{\mathbf{Y}}^{k-1}) = h_i(\mathbf{S}^{k-1})$ depending on the vector $\mathbf{S}^{k-1} = \mathbf{A}^{k-1} \mathbf{Y}^{k-1}$ of counts of infected contacts for each student at week $k - 1$.

Simulation Setup

Three scenarios:

Null	Formation Model	Persistence Model	Infection Probability
Hypothesis	Parameters θ^+	Parameters θ^-	Function h_i
$H_0^\sharp$	$(-0.5, 0, 0, 0)$	$(-0.5, 0, 0, 0)$	$h_i(\mathbf{s}) = 0.5$
$\overline{H_0^A} \cap H_0^Y$	$(-0.2, 0, 0, -1)$	$(-0.2, 0, 0, -1)$	$h_i(\mathbf{s}) = 0.5$
$\overline{H_0^A} \cap \overline{H_0^Y}$	$(-0.2, 0, 0, -1)$	$(-0.2, 0, 0, -1)$	$h_i(\mathbf{s}) = \frac{n}{2} \text{expit}(s_i - \bar{s}) / \sum_{j=1}^n \text{expit}(s_j - \bar{s})$

Table: The null hypothesis refers to the null hypothesis that is true under the data generating process, formation model parameters are $\theta^+ = (\theta_{\text{edges}}^+, \theta_Z^+, \theta_Y^+, \theta_{ZY}^+)$, persistence model parameters are $\theta^- = (\theta_{\text{edges}}^-, \theta_Z^-, \theta_Y^-, \theta_{ZY}^-)$, the infection probability function $h_i(\mathbf{s})$ gives the probability of student i being infected at week k given the vector $\mathbf{S}^{k-1} = \mathbf{s} = (s_1, \dots, s_n)$ of infected contact counts for each student at week $k-1$, and $\bar{s} = n^{-1} \sum_{i=1}^n s_i$.

Three p-values:

- $\rho_B^\sharp$: testing the sharp null
- ρ_B^Y : testing H_0^Y using known q
- $\hat{\rho}_B^Y$: testing H_0^Y using estimated q

Simulation Study

Goal: use simulated data to investigate the performance of our testing procedure

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- $n \in \{24, 48\}$ students
- $\tau \in \{5, 10\}$ weeks
- 50:50 cluster randomization with 12 equal-sized clusters

Simulation Study

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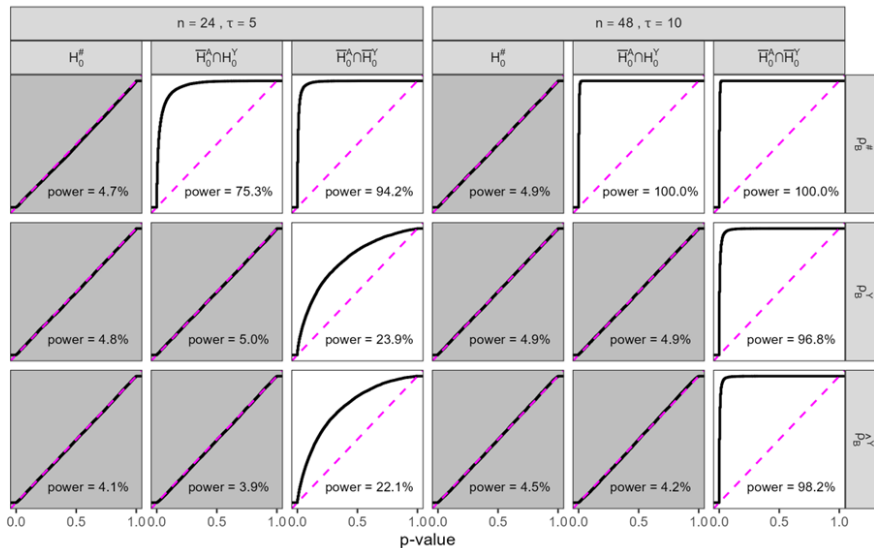
- $n \in \{24, 48\}$ students
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- 50:50 cluster randomization with 12 equal-sized clusters
- Three scenarios:
 - 1 $H_0^\sharp$: no effect of intervention on networks or infection
 - 2 $\overline{H}_0^A \cap H_0^Y$: no effect of intervention on infection
 - 3 $\overline{H}_0^A \cap \overline{H}_0^Y$: effect of intervention on both networks and infection

Simulation Study

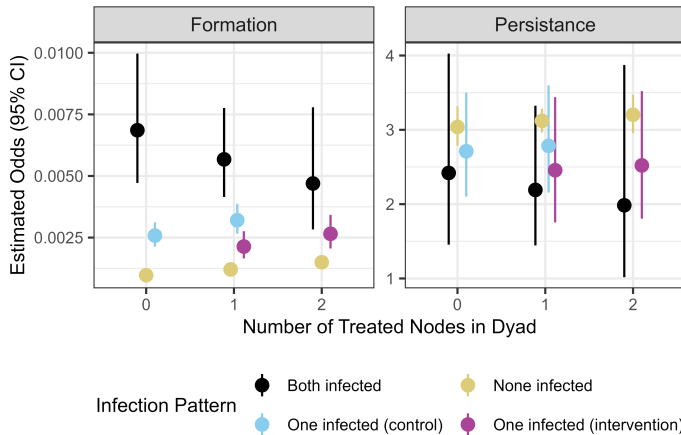
Goal: use simulated data to investigate the performance of our testing procedure

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 - 3 $\overline{H}_0^A \cap \overline{H}_0^Y$: effect of intervention on both networks and infection
- Three testing procedures:
 - 1 $\rho_B^\#$: testing $H_0^\#$
 - 2 ρ_B^Y : testing H_0^Y using known q
 - 3 $\hat{\rho}_B^Y$: testing H_0^Y using estimated q

Simulation Results: Empirical CDF of P-Values



eX-FLU: STERGM Results



Type I Error Control

- **Proposition 2.1:** Let T_N be a test statistic with CDF F_N under H_0 .
 - ① Under H_0 , $F_N(T_N)$ stochastically dominates a $\text{Uniform}(0, 1)$ distribution for any N .
 - ② If the test statistic T_N has a continuous limiting distribution, then $F_N(T_N) \rightarrow^d \text{Uniform}(0, 1)$.
- **Corollary 2.2:**
 - ① Under $H_0^\sharp$, the sharp null p-value $\rho_N^\sharp$ stochastically dominates a $\text{Uniform}(0, 1)$ distribution for any N .
 - ② Under H_0^Y , the oracle p-value ρ_N^Y stochastically dominates a $\text{Uniform}(0, 1)$ distribution for any N .
 - ③ If the test statistic T_N has a continuous limiting distribution, then $\rho_N^\sharp \rightarrow^d \text{Uniform}(0, 1)$ under $H_0^\sharp$ and $\rho_N^Y \rightarrow^d \text{Uniform}(0, 1)$ under H_0^Y .

Type I Error Control

- **Proposition 2.3:** Let $q(\mathbf{a}, \mathbf{z}, \theta) \equiv \Pr\{\mathbf{A}(\mathbf{z}) = \mathbf{a}; \theta\}$ denote the PMF of the distribution of stochastic potential networks $\mathbf{A}(\mathbf{z})$ at parameter value θ , let $\hat{\theta}_N$ denote the estimator of θ , and let $F_N(\cdot; \theta)$ denote the CDF of the test statistic T_N at θ , with limiting CDF $F(\cdot; \theta)$. Let θ_0 denote the true value of θ . Assume the following:

(A1) $\hat{\theta}_N \rightarrow^P \theta_0$

(A2) $F(t; \theta_0)$ is continuous in t on $\mathbb{R}$

(A3) there exists a $\delta_0 > 0$ such that

$$\sup_{\theta \in B_{\delta_0}(\theta_0)} \sup_{t \in \mathbb{R}} |F_N(t; \theta) - F(t; \theta)| \rightarrow 0$$

(A4) $F(t; \theta)$ is continuous in θ at θ_0 uniformly in t , i.e.,

$$\lim_{\theta \rightarrow \theta_0} \sup_{t \in \mathbb{R}} |F(t; \theta) - F(t; \theta_0)| = 0$$

Then the plug-in p-value $\hat{\rho}_N^Y$ converges in distribution to $\text{Uniform}(0, 1)$.

Type I Error Control

- **Proposition 2.4:** Let $\rho_N = F_N(T_N)$ for test statistic T_N and CDF F_N (not necessarily the true CDF of T_N). Let $T_N^* = h_N(T_N)$ for a sequence of deterministic, strictly increasing functions h_N . Define $F_N^*(t) = F_N\{h_N^{-1}(t)\}$, the (not necessarily true) CDF of the transformed test statistic, and let $\rho_N^* = F_N^*(T_N^*)$. Then $\rho_N^* = \rho_N$.
- **Corollary 2.5:** Let h_N be a sequence of deterministic, strictly increasing functions.
 - 1 If the hypotheses of Proposition 2.1 are met for a test statistic T_N , then the results also hold for $T_N^* = h_N(T_N)$.
 - 2 If the hypotheses of Proposition 2.3 are met for $T_N, F_N(\cdot; \theta)$, then the results also hold for $T_N^* = h_N(T_N), F_N^*(\cdot; \theta) = F_N\{h_N^{-1}(\cdot); \theta\}$.

Supplement to Project 3 : Causal Inference from Cluster Randomized Trials with Differential Nonresponse

Brian Richardson, Bonnie Shook-Sa, Michael Hudgens

Simulation Details

- $m = 20$ clusters
 - ▶ cluster treatment $\Pr(A = 1) = 0.5$
 - ▶ cluster covariate $X \sim N(0, 1)$
- invited trial sample of size $n^{\text{trial}} \in \{500, 5000\}$
- individual covariates $W_1 \sim \text{Bernoulli}(0.5)$, $W_2 \sim N(0, 1)$
- potential outcomes

$$Y(0) \sim \text{Bernoulli}\{\text{expit}(-0.5 + 0.25X_1 + 0.5W_1 + 0.5W_2)\},$$

$$Y(1) \sim \text{Bernoulli}\{\text{expit}(0 - 1W_1 - 0.5W_2)\}$$

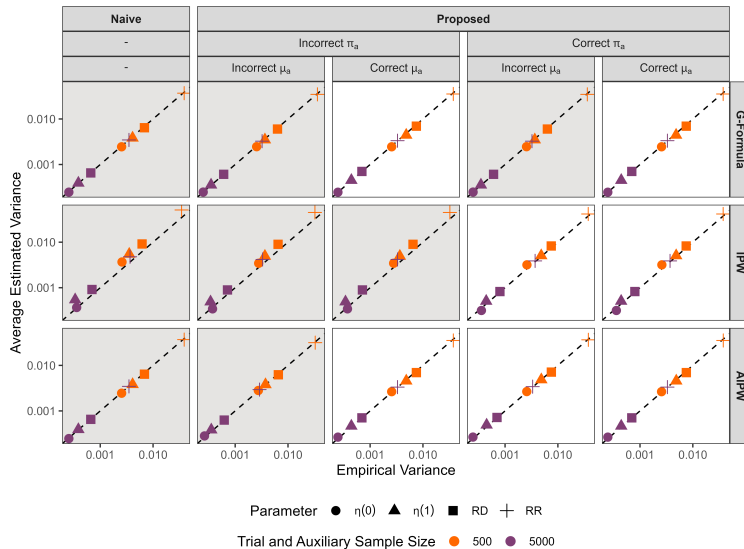
$\implies$ treatment is marginally protective, but harmful when $W_1 = 0$

- observed outcomes $Y = (1 - A)Y(0) + AY(1)$
- response $R \sim \text{Bernoulli}\{\text{expit}(\alpha_0 - 2AW_1)\}$
 - $\implies$ marginal $\Pr(R = 1) = 0.5$, but higher response rate if $W_1 = 0$
- auxiliary sample of size $n^{\text{aux}} \in \{500, 5000\}$
 - ▶ same covariates W_1, W_2

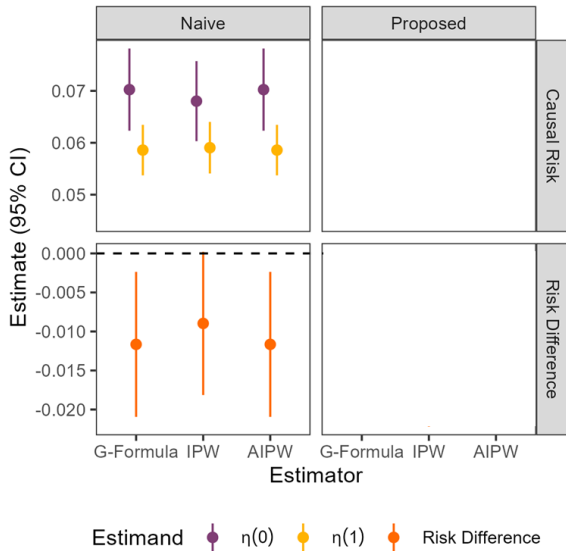
Simulation Details

- naive and proposed versions of the g-formula, IPW, and AIPW estimators of $\eta(a)$
 - ▶ naive versions used trial data only, and ignored nonresponse
- Estimand: $RD = \eta(1) - \eta(0)$
- Estimator: $\widehat{RD} = \widehat{\eta}(1) - \widehat{\eta}(0)$
 - ▶ variance estimated using the delta method
- Outcome model specification
 - ▶ μ_a correctly modeled with logistic regression with covariates X, W_1, W_2 and their interaction with treatment A
 - ▶ μ_a mis-specified by excluding the covariate W_1 and its interaction with A
- Joint propensity score model specification
 - ▶ censoring mechanism π_a^C always correctly specified as logistic regression with covariates X, W_1, W_2 and their interaction with treatment A
 - ▶ response mechanism π_a^{AR} possibly mis-specified by incorrectly assuming $R_{ij} \perp\!\!\!\perp A_i | \mathbf{X}_i, \mathbf{W}_{ij}$

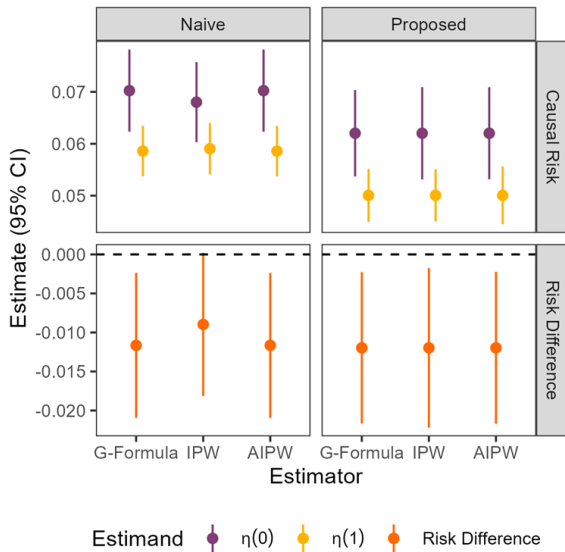
Simulation: Variance Estimation



Supplemental Analysis: Arms A/B vs Standard of Care



Supplemental Analysis: Arms A/B vs Standard of Care



Supplement to Project 1 : Addressing Confounding and Continuous Exposure Measurement Error Using Corrected Score Functions

Brian Richardson, Brian Blette, Peter Gilbert, Michael Hudgens

Corrected Score Functions: What?

- Suppose the oracle score function is **conditionally unbiased**, meaning

$$E\{\Psi_0(Y, \mathbf{L}, \mathbf{A}; \theta) | \mathbf{A}\} = \mathbf{0}.$$

- Define the corrected score function as

$$\Psi_{CS}(Y, \mathbf{L}, \mathbf{A}^*; \theta) = E \left[\text{Re} \left\{ \Psi_0(Y, \mathbf{L}, \tilde{\mathbf{A}}; \theta) \right\} \mid Y, \mathbf{L}, \mathbf{A}^* \right],$$

where $\tilde{\mathbf{A}} = \mathbf{A}^* + i\tilde{\epsilon}$, $i = \sqrt{-1}$, $\text{Re}(\cdot)$ denotes the real component of a complex number, and $\tilde{\epsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma_{me})$.

- Then

$$\begin{aligned} E\{\Psi_{CS}(Y, \mathbf{L}, \mathbf{A}^*; \theta) \mid Y, \mathbf{L}, \mathbf{A}\} &= \Psi_0(Y, \mathbf{L}, \mathbf{A}; \theta) \\ \implies E[E\{\Psi_{CS}(Y, \mathbf{L}, \mathbf{A}^*; \theta) \mid Y, \mathbf{L}, \mathbf{A}\}] &= E\{\Psi_0(Y, \mathbf{L}, \mathbf{A}; \theta)\} \\ \implies E\{\Psi_{CS}(Y, \mathbf{L}, \mathbf{A}^*; \theta)\} &= \mathbf{0} \end{aligned}$$

Corrected Score Functions: Why?

The key result ([Novick and Stefanski, 2002](#)) for corrected score functions is that, for a smooth enough function $\mathbf{f}$, the function $\tilde{\mathbf{f}}$ defined by

$$\tilde{\mathbf{f}}(\mathbf{A}^*) \equiv \mathbb{E}[\text{Re}\{\mathbf{f}(\mathbf{A}^* + i\tilde{\epsilon})\} | \mathbf{A}, \mathbf{A}^*]$$

does not depend on $\mathbf{A}$ and satisfies

$$\mathbb{E}\{\tilde{\mathbf{f}}(\mathbf{A}^*) | \mathbf{A}\} = \mathbf{f}(\mathbf{A}).$$

The proof of this for the special case $f(\mathbf{A}) = \exp(\mathbf{c}\mathbf{A}^T)$ is illustrative.

Corrected Score Functions: Why?

CLAIM: $E\{\tilde{f}(\mathbf{A}^*)|\mathbf{A}\} = f(\mathbf{A})$ for $f(\mathbf{A}) = \exp(\mathbf{c}\mathbf{A}^T)$.

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$$\begin{aligned}\tilde{f}(\mathbf{A}^*) &\equiv E(\text{Re}[\exp\{\mathbf{c}(\mathbf{A}^* + i\tilde{\epsilon})^T\}]|\mathbf{A}, \mathbf{A}^*) \\ &= \exp(\mathbf{c}\mathbf{A}^{*T}) \underbrace{\text{Re}[E\{\exp(i\mathbf{c}\tilde{\epsilon}^T)\}]}_{\text{normal c.f.}} = \exp(\mathbf{c}\mathbf{A}^{*T}) \exp\left(-\frac{1}{2}\mathbf{c}\Sigma_{me}\mathbf{c}^T\right)\end{aligned}$$

Corrected Score Functions: Why?

CLAIM: $E\{\tilde{f}(\mathbf{A}^*)|\mathbf{A}\} = f(\mathbf{A})$ for $f(\mathbf{A}) = \exp(\mathbf{c}\mathbf{A}^T)$.

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